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HCM-UTE

TUTORIAL

Single-Stage Amplifiers

Common-Source, Source-Follower, Common-Gate, and Cascode Stages

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

Based on Design of Analog CMOS Integrated Circuits, Chapter 3 (Razavi)

OUTLINE

Roadmap for This Lecture

Eight sections, from device fundamentals to cascode stages

I Introduction & Review

- Why analog design is a multi-dimensional trade-off
- AC ground, Thévenin/Norton, and load-line review

II Common-Source Amplifier

- Small-signal model and gain derivation
- Effect of channel-length modulation (ro)

III CS Amplifier with Practical Loads

- Diode-connected NMOS and PMOS loads
- Current-source load

IV CS Amplifier with Source Degeneration

- Effective transconductance Gm and output resistance Ro
- Linearity vs. gain trade-off

V Output Resistance of the CS Stage

- Test-source derivation of Rout
- Physical intuition for the RS boost

VI Source Follower (Common Drain)

- Gm and Rout derivation
- Three key disadvantages

VII Common-Gate Amplifier

- Gm and Rout derivation
- Input resistance and impedance matching

VIII Cascode & Folded-Cascode

- Rout boosting via cascoding
- Current-source mismatch suppression



Introduction & Review

- Why analog design is a multi-dimensional trade-off
- AC ground, Thévenin/Norton, and load-line review

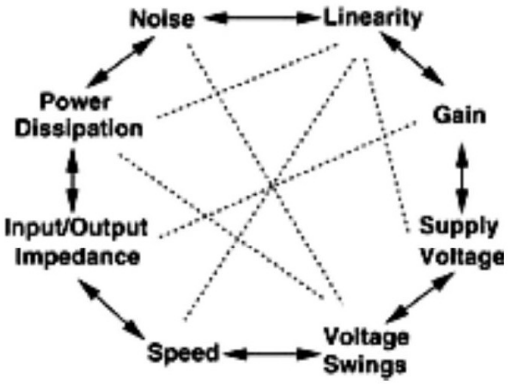
What Makes Analog Design Hard?

- Circuits map input signals through a transistor's inherently nonlinear I–V characteristic
- Design requires balancing power, gain, bandwidth, linearity, voltage swing, and impedance simultaneously
- These competing trade-offs motivate the systematic single-stage amplifier study in this lecture



Non-linear system

Trade-offs

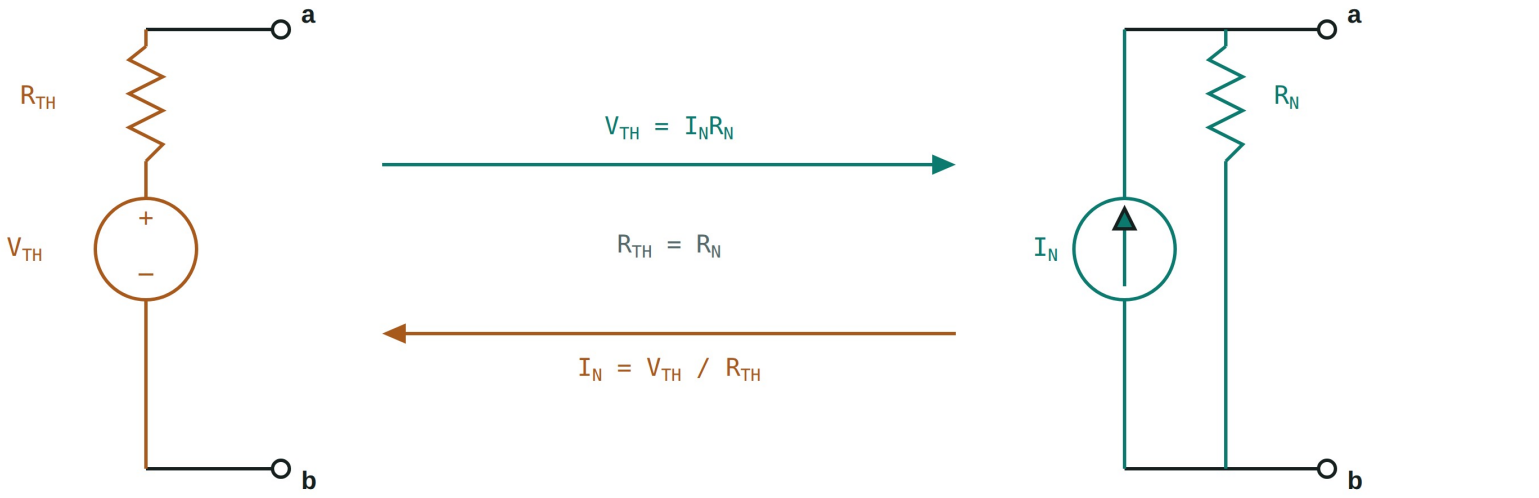


Multi-dimensional optimization

Circuit Analysis Toolkit Review (*)

- AC ground removes DC supplies from the small-signal circuit
- Thévenin's and Norton's equivalents simplify source and load networks
- Load-line analysis and the large-signal vs. small-signal distinction set up the derivations ahead

- Ac ground
- Thevenin's circuit & Norton's circuit
- Load line
- small signal and large signal



Thévenin element (series branch)

Norton element (parallel branch)

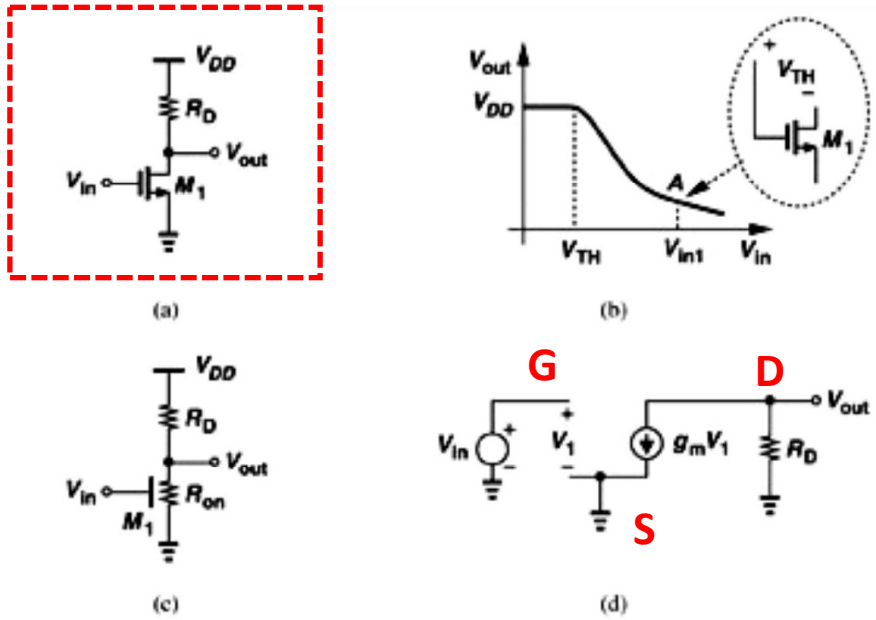
II

Common-Source Amplifier

- Small-signal model and gain derivation
- Effect of channel-length modulation (ro)
- Load-line interpretation of gain

CS Stage: Small-Signal Model in Saturation

- The small-signal model here neglects channel-length modulation as a first pass
- Vin sets the operating point that separates the saturation and triode regions of M1
- This model is the starting point for deriving the CS voltage gain



Small signal model for the saturation region (channel length modulation neglected)

Vin is a transition point between saturation and linear region

Large- and Small-Signal Analysis (*)

- Large-signal Vout is found separately in the linear/triode and saturation regions
- Small-signal gain Av = ∂Vout/∂Vin = −RD·gm, independent of the large-signal expression's region
- gm = μnCox(W/L)(Vin−VTH) links the bias point directly to the achievable gain

➢Large signal analysis

➢Channel length modulation is neglected

$$\text{Linear Region}: V_{out} = V_{DD} - I_D R_D = V_{DD} - R_D \mu_n C_{OX} \frac{W}{L}\left[(V_{in} - V_{TH})V_{out} - \frac{1}{2}V_{out}^{\ 2}\right]$$

$$\text{Saturation Region}: V_{out} = V_{DD} - I_D R_D = V_{DD} - R_D \frac{1}{2}\mu_n C_{OX} \frac{W}{L}(V_{in} - V_{TH})^2$$

➢Small signal analysis

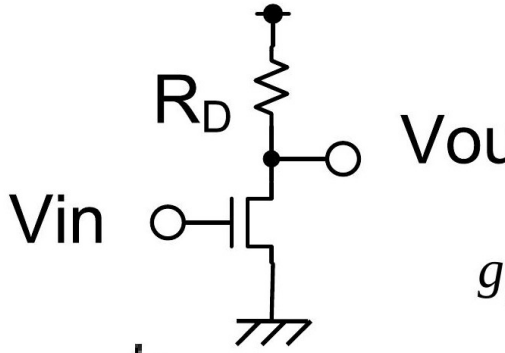
$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = -R_D \mu_n C_{OX} \frac{W}{L}(V_{in} - V_{TH}) = -R_D \cdot g_m$$

$$\because g_m = \mu_n C_{OX} \frac{W}{L}(V_{in} - V_{TH})$$

Ex. 3.1 — Sketching ID and gm vs. Vin

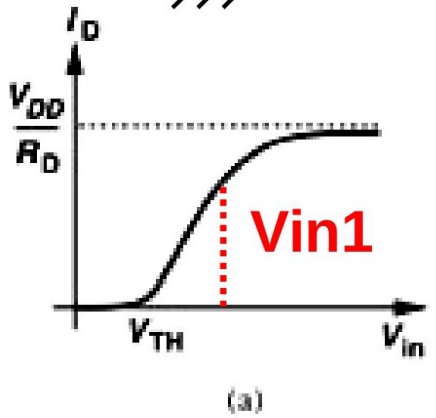
- In saturation, gm grows with (Vin−VTH); in the triode region, gm becomes roughly constant
- The Vin1 marker highlights the saturation-to-linear transition point
- Sketching ID and gm together builds intuition for how the bias point sets the gain

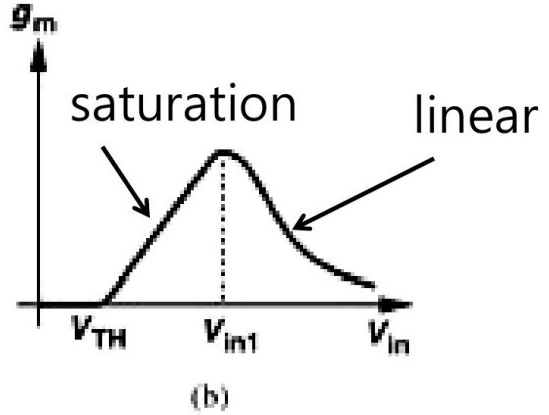
Ex3.1) sketch the drain current and gm



(Saturation g_m, Linear g_m)

$$g_m = \mu_n C_{OX} \frac{W}{L}(V_{in} - V_{TH}), g_m = \mu_n C_{OX} \frac{W}{L} V_{DS}$$

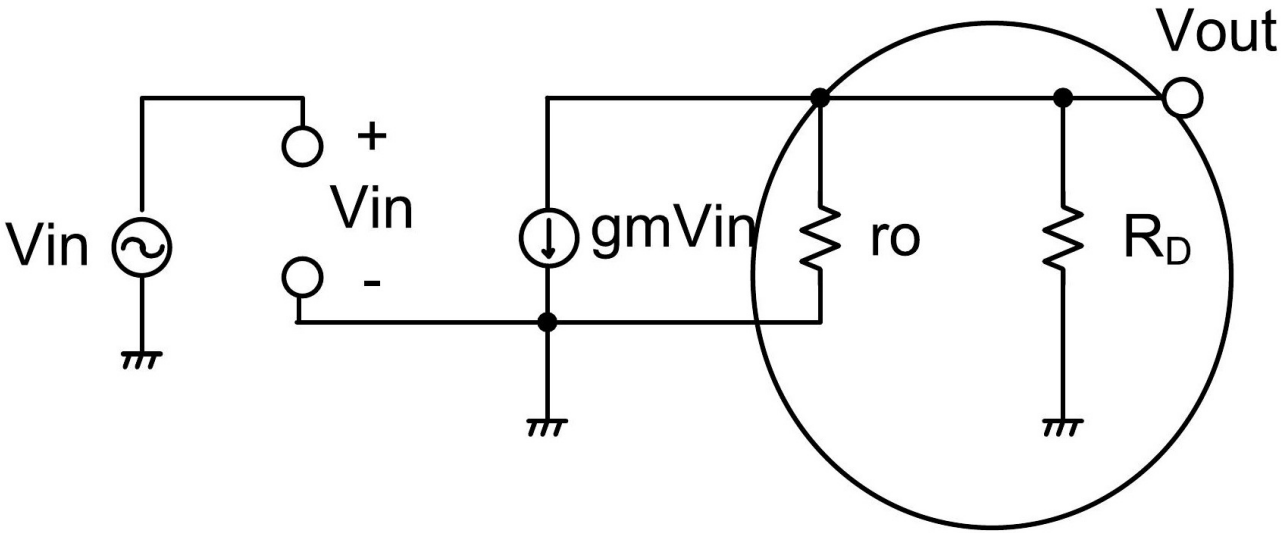




Including Channel-Length Modulation (ro)

- Adding ro in parallel with RD gives Av = −gm(ro∥RD)
- As RD → ∞, the gain approaches −gm·ro — the transistor's intrinsic gain
- Channel-length modulation therefore caps the maximum achievable gain

➢If we consider channel length modulation (ro),

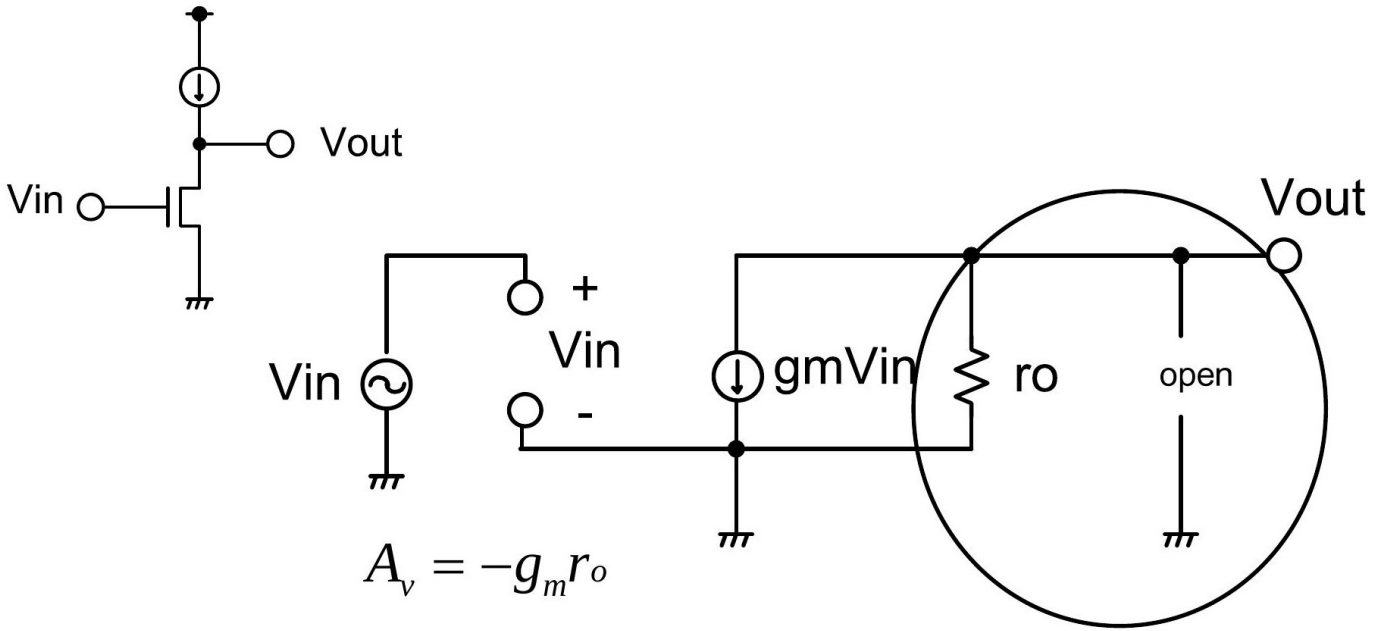


$$A_v = -g_m(r_o \| R_D)$$

Ex. 3.2 — Voltage Gain with ro (Intrinsic Gain)

- With RD replaced by an open circuit, Av = −gm·ro, the intrinsic-gain limit
- This example isolates the transistor's own gain ceiling, independent of any external load

Ex3.2) Calculated the small-signal voltage gain

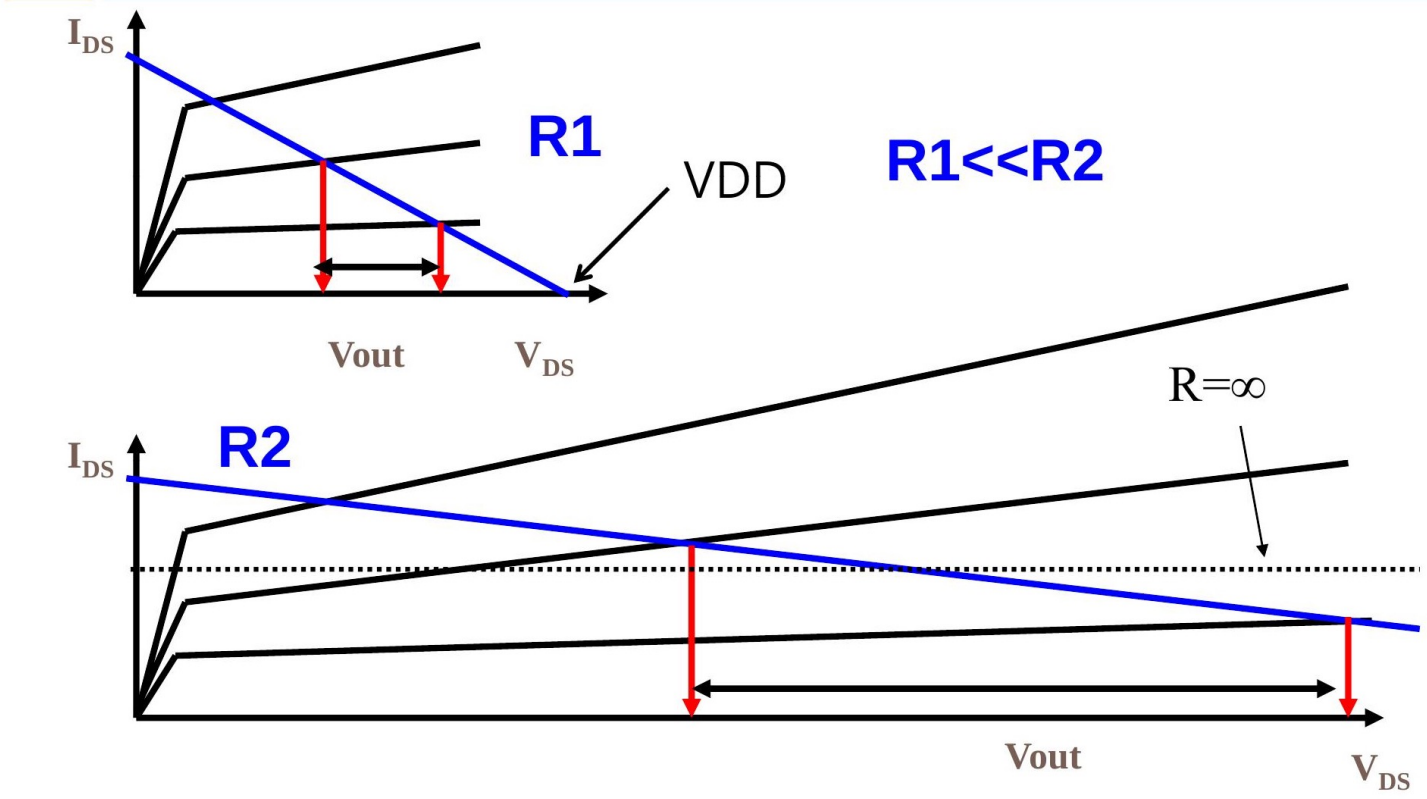


$$A_v = -g_m r_o$$

" *intrinsic gain*"

Load-Line View of Gain vs. Load Resistance

- A steeper load line (small R, R1) spans a narrow ID range and gives a smaller swing for a given gm
- A shallower load line (large R, R2) approaches the R = ∞ intrinsic-gain case
- This graphical view ties the load-line slope directly to the achievable voltage gain



III

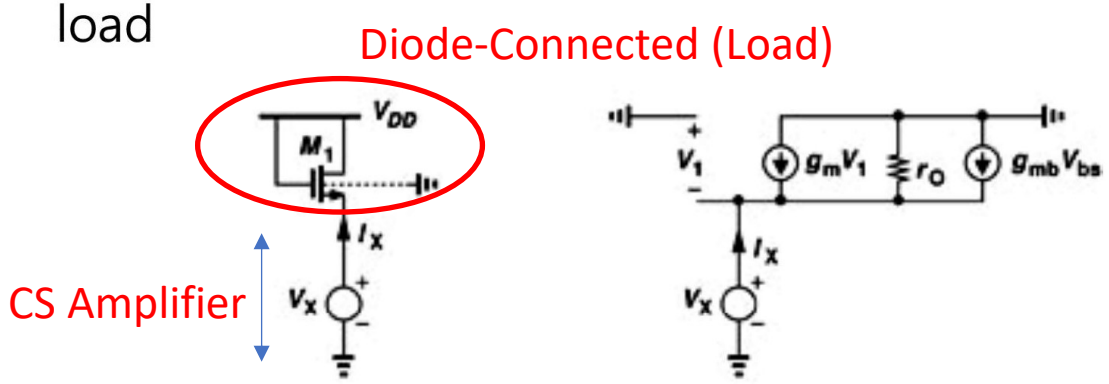
CS Amplifier with Practical Loads

- Diode-connected NMOS and PMOS loads
- Current-source load
- Triode-region load

Diode-Connected Load — Thévenin Equivalent

- A diode-connected device (gate tied to drain) presents an equivalent resistance at its Vx/Ix terminals
- KCL at that node: (gm+gmb)Vx + Vx/ro = Ix
- Solving gives Vx/Ix ≈ 1/(gm+gmb), i.e. approximately 1/gm when body effect is neglected

Diode-connected load is seen from Vx and IX
In other words, get Thevenin's circuit of diode-connected load

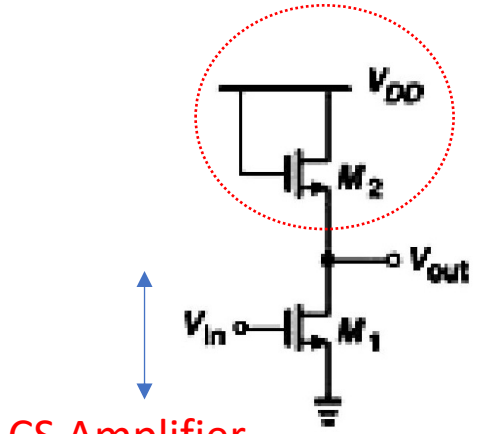


$$(g_m + g_{mb})V_x + \frac{V_x}{r_o} = I_x$$

$$\frac{V_x}{I_x} = \frac{1}{g_m + g_{mb}} \| r_o \approx \frac{1}{g_m + g_{mb}} \approx \frac{1}{g_m}$$

Voltage Gain with a Diode-Connected Load

- Av = −gm1(1/(gm2+gmb2) ∥ ro1) ≈ −(gm1/gm2)·1/(1+η)
- The gain reduces to −√[(W/L)1/(W/L)2] — set purely by the transistor sizing ratio
- Because bias current cancels out, this gain is independent of the bias current — a key practical advantage



CS Amplifier

$$A_v = -g_{m1}\left(\frac{1}{g_{m2}+g_{mb2}} \| r_{o1}\right) \approx -\frac{g_{m1}}{g_{m2}}\frac{1}{1+\eta}$$

$$A_v = -\sqrt{\frac{(W/L)_1}{(W/L)_2}}\frac{1}{1+\eta}$$

Gain is independent of bias current!

Sizing Example — Achieving Av = 10

- Av = −10 requires (W/L)1/(W/L)2 = 100, i.e. W1 must be 100× larger than W2
- This large ratio also makes (VGS2−VTH2) about 10× larger than (VGS1−VTH1)
- Maintaining equal bias current through such mismatched devices creates a DC biasing challenge

$$A_v = -\sqrt{\frac{(W/L)_1}{(W/L)_2}} = -10$$

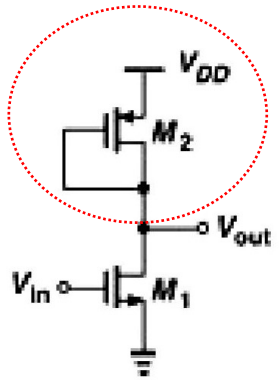
$$\sqrt{\frac{(W/L)_1}{(W/L)_2}} = 10 \Rightarrow \frac{(W/L)_1}{(W/L)_2} = 100$$

➢If Av=10, W1 is 100 times larger than W2.
➢If W1 is 100 times larger than W2, VGS2-VTH is 10 times large than VGS1-VTH, to maintain the same current through W1 and W2
➢This causes DC biasing problem

PMOS Diode-Connected Load

- Av = −√[μn(W/L)1 / (μp(W/L)2)], since μn = μpCox/2 differs from the NMOS case only by mobility ratio
- A PMOS load's body can be tied to its own source, eliminating body effect and gmb entirely

➢PMOS load



$$A_v \approx -\sqrt{\frac{u_n (W/L)_1}{u_p (W/L)_2}}$$

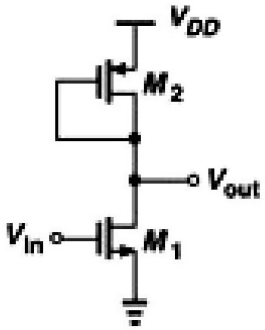
$$\because u_n = \frac{u_p C_{OX}}{2}$$

What is different between NMOS and PMOS diode?

No body effect => gmb eliminated

Gain Sensitivity to Device Mismatch

- Equating ID1 = ID2 for the NMOS input device and the PMOS load relates gm1/gm2 to the overdrive-voltage ratio
- Av = −|VGS2−VTH2| / |VGS1−VTH1|, so gain depends directly on that ratio
- This dependence implies a substantial voltage-swing / headroom constraint in practice



$$I_{D1} = I_{D2}$$

$$u_n\left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH1})^2 = u_p\left(\frac{W}{L}\right)_2 (V_{GS2} - V_{TH2})^2$$

$$A_v = -\frac{gm1}{gm2} = -\sqrt{\frac{u_n (W/L)_1}{u_p (W/L)_2}} = -\frac{|V_{GS2} - V_{TH2}|}{|V_{GS1} - V_{TH1}|}$$

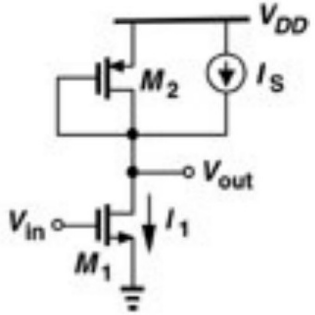
$$= -\frac{u_n (W/L)_1 (|V_{GS1} - V_{TH1}|)}{u_p (W/L)_2 (|V_{GS2} - V_{TH2}|)}$$

This implies substantial voltage swing constraint.

Ex. 3.3 — Gain with Unequal Bias Currents

- When the load current is only 0.75 of the input branch current (IS = 0.75·I1), the gain expression must be re-derived
- Result: |VGS2−VTH2| / |VGS1−VTH1| = Av/4, linking the biasing ratio directly to the realized gain

Ex3.3) when $I_S=0.75I_1$, find Av



$$A_v = -\frac{gm1}{gm2} = -\sqrt{\frac{4u_n(W/L)_1}{u_p(W/L)_2}}$$

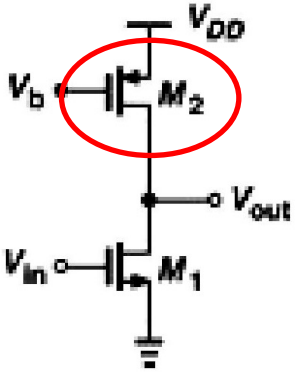
$$u_n(W/L)_1(VGS_1 - VTH_1)^2 = 4u_p(W/L)_2(VGS_2 - VTH_2)^2$$

$$\frac{|VGS_2 - VTH_2|}{VGS_1 - VTH_1} = \sqrt{\frac{u_n(W/L)_1}{4u_p(W/L)_2}} = \frac{A_v}{4}$$

Current-Source Load

- Replacing the resistive/diode load with an ideal current source gives Av = −gm1(ro1∥ro2)
- Assuming ro2 is large, Av ≈ −gm1·ro1 = −√(2μnCoxID·W/L)·1/(λID)
- This load approaches the intrinsic-gain limit without the extra headroom a resistor would consume

Ideal current source



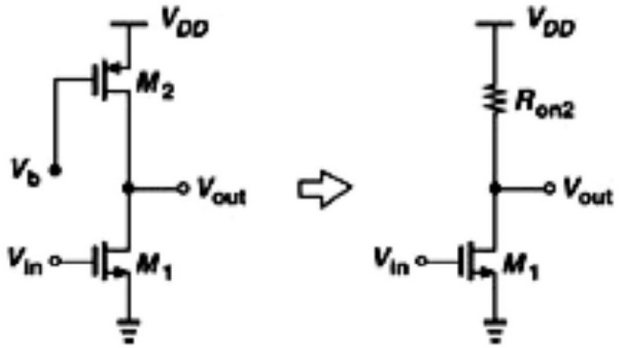
$$A_v = -g_{m1}(r_{o1} \| r_{o2})$$

Assuming r_{o2} large,

$$A_v = -g_m r_{o1} = -\sqrt{\left[2\mu_n C_{ox} I_D \left(\frac{W}{L}\right)_1\right]} \frac{1}{\lambda I_D}$$

Triode-Region (Linear) Load

- Biasing M2 deep in the triode region requires Vout−Vb ≥ |VTHP|
- In this region M2 behaves as a linear resistor: RON2 = 1/[μpCox(W/L)2(VDD−Vb−|VTHP|)]
- The resulting gain Av = −gm·RON2 lets the effective load be tuned via the gate bias Vb



$$V_{out} - V_b \geq |V_{THP}| \Rightarrow \text{Linear Region}$$

$$A_v = -g_m R_{ON2}$$

$$R_{ON2} = \frac{1}{\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{DD} - V_b - |V_{THP}|)}$$

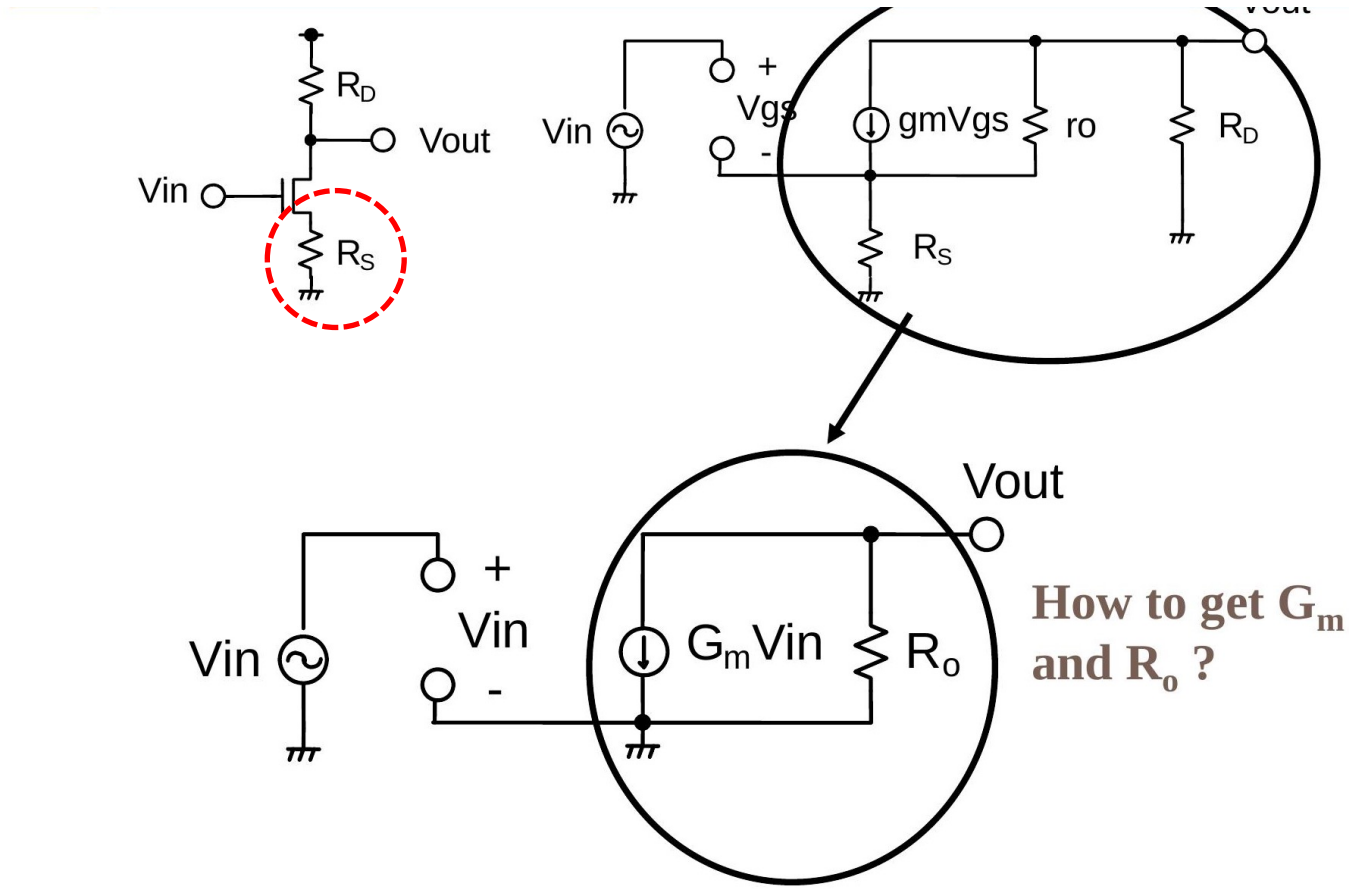
IV

CS Amplifier with Source Degeneration

- Effective transconductance Gm and output resistance Ro
- Linearity vs. gain trade-off
- Worked design examples

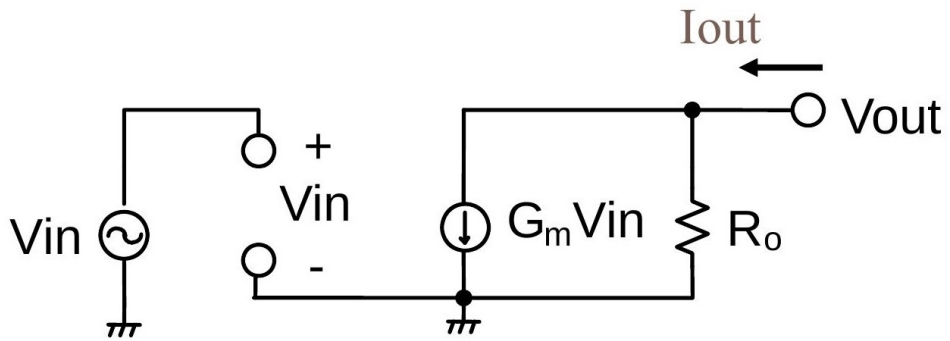
Adding Source Degeneration — Overview

- A resistor RS in the source path trades some voltage gain for improved linearity and predictability
- The stage is analyzed through an **effective transconductance Gm** and **effective output resistance Ro**
- These two parameters fully characterize the degenerated stage's small-signal behavior



Defining Gm and Ro by Test Sources

- Gm ≡ Iout/Vin, measured with Vout shorted to 0 V; Ro ≡ Vout/Iout, measured with Vin shorted to 0 V
- These two test-source definitions decouple the gain and output-resistance calculations
- Together they give Vout = −Gm·Vin·Ro for the loaded stage



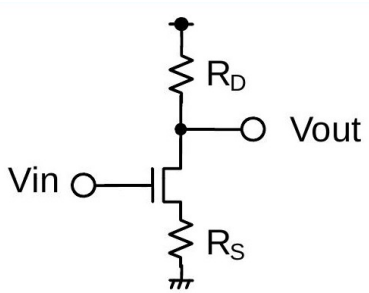
$$G_m = I_{out} / V_{in} \quad when \ V_{out} = 0V$$

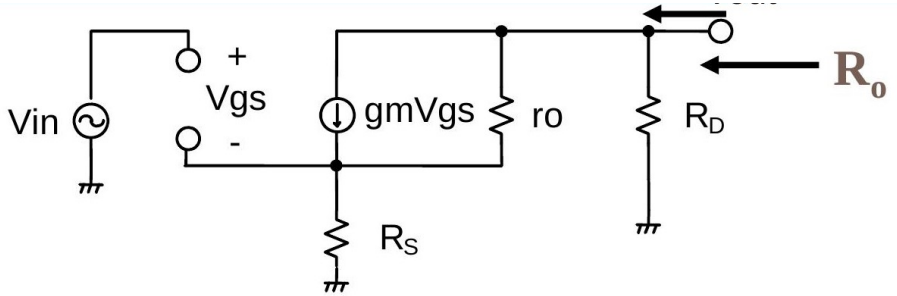
$$R_o = V_{out} / I_{out} \quad when \ V_{in} = 0V$$

$$\therefore V_{out} = -G_m V_{in} R_o$$

Gm with ro → ∞ (Simplified Case)

- Setting ro = ∞ gives Gm = gm/(1+gmRS), a reduced effective transconductance
- With Ro = RD in this limit, Av = −Gm·RD = −gm·RD/(1+gmRS)
- Source degeneration lowers gain by the same (1+gmRS) factor that improves linearity





$set\ r_o = \infty$

$$I_{out} = g_m V_{gs}, V_{in} = g_m V_{gs} R_S + V_{gs}$$

$$G_m = I_{out}/V_{in} = g_m V_{gs}/(g_m V_{gs} R_S + V_{gs}) = g_m/(1+g_m R_S)$$

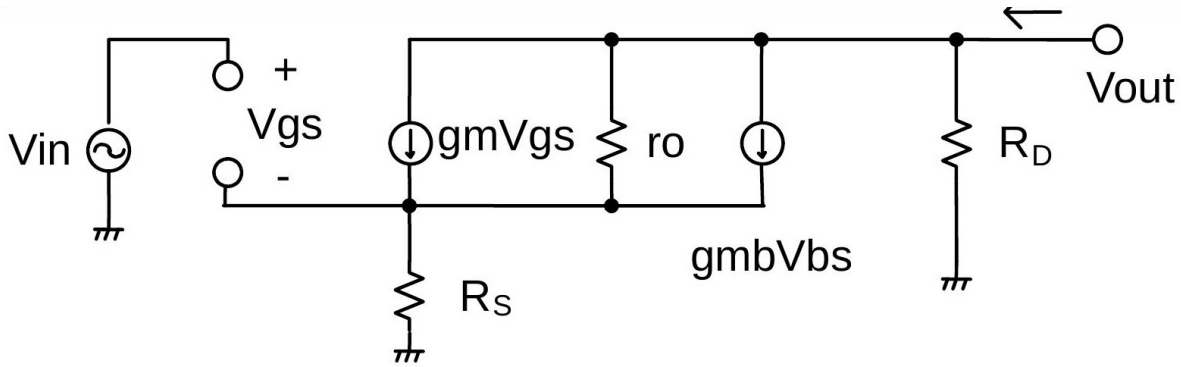
g_m => G_m

Ro=R_D when r_o is infinite

$$\therefore A_v = -G_m R_o = -\frac{g_m}{1+g_m R_S} R_D$$

General Gm — Including gmb and ro

- With body effect and finite ro included, Gm = gm·ro / {RS[1+(gm+gmb)RS]+ro} — found by setting Vout = 0
- This is the full expression that reduces to the simplified case as ro → ∞

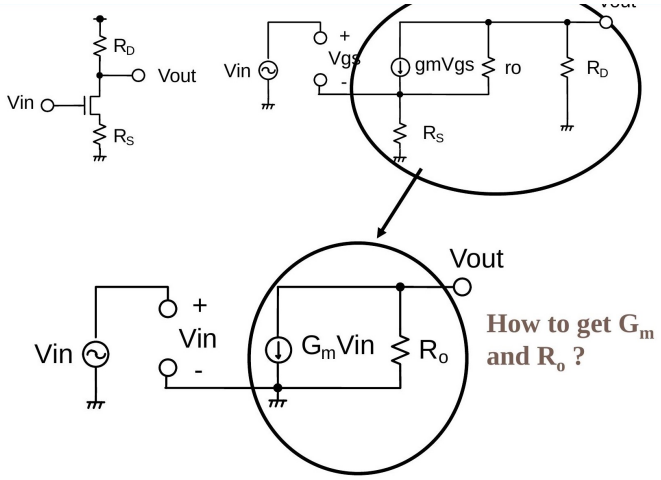


➢**gmb and ro considered**

➢**Set Vout=0V**

$$G_m = \frac{g_m r_o}{R_S + [1 + (g_m + g_{mb}) R_S] r_o}$$

Step-by-Step Gm Derivation (***)



- Writing Iout in terms of Vgs and Vbs and substituting the RS feedback relation gives a self-consistent loop equation
- Solving for Gm = Iout/Vin yields Gm = gm / (1+gmRS+gmbRS+RS/ro)
- Every degeneration and body-effect term appears explicitly in the denominator

$$I_{out} = g_m V_{gs} + g_{mb} V_{bs} - \frac{I_{out} R_S}{r_o}$$

$$I_{out} = g_m (V_{in} - I_{out} R_S) + g_{mb} (-I_{out} R_S) - \frac{I_{out} R_S}{r_o}$$

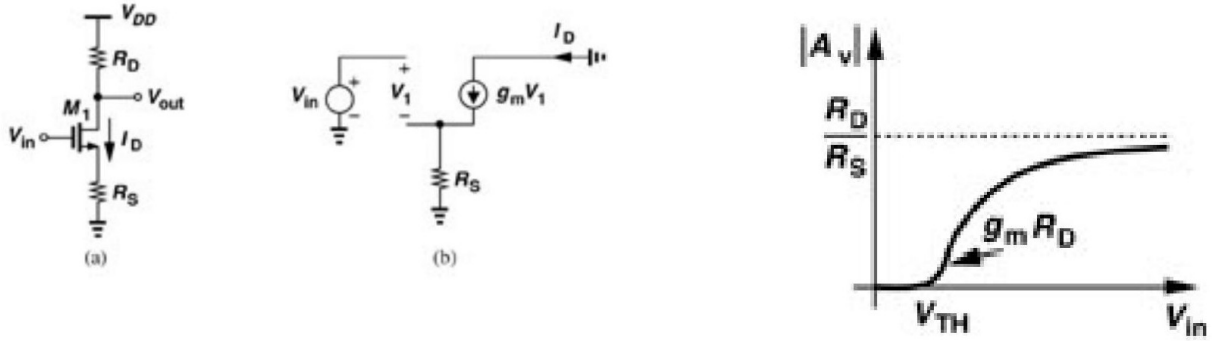
$$I_{out} \left[1 + g_m R_S + g_{mb} R_S + \frac{R_S}{r_o} \right] = g_m V_{in}$$

$$G_m = I_{out} / V_{in} = \frac{g_m}{1 + g_m R_S + g_{mb} R_S + \frac{R_S}{r_o}}$$

Ex. 3.4 — Gain vs. Vin Characteristic (***)

- As |Av| is plotted against Vin, the curve flattens because Av → gmRD/(1+gmRS) for large Vin
- For gmRS ≫ 1, this asymptote simplifies to RD/RS — gain is then set almost entirely by a resistor ratio

Ex3.4) Plot the small-signal voltage gain of the circuit



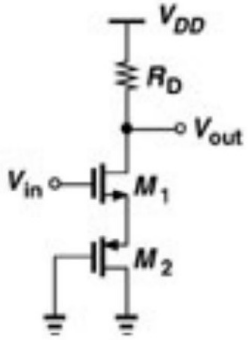
As the Vin becomes larger, Av becomes RD/RS

$$Av = \frac{g_m R_D}{1 + g_m R_S} \Rightarrow \frac{R_D}{R_S}$$

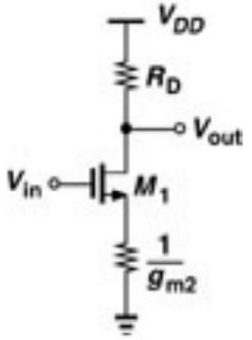
Ex. 3.5 — Ideal Case, No CLM or Body Effect

- With channel-length modulation and body effect both neglected, Av reduces to RD / (1/gm1 + 1/gm2)
- This simplified result cleanly separates the roles of the amplifying and degenerating devices

Ex3.5) Assume channel length modulation=0, body effect=0, calculate the voltage gain



(a)



(b)

$$A_v = \frac{R_D}{\frac{1}{g_{m1}} + \frac{1}{g_{m2}}}$$

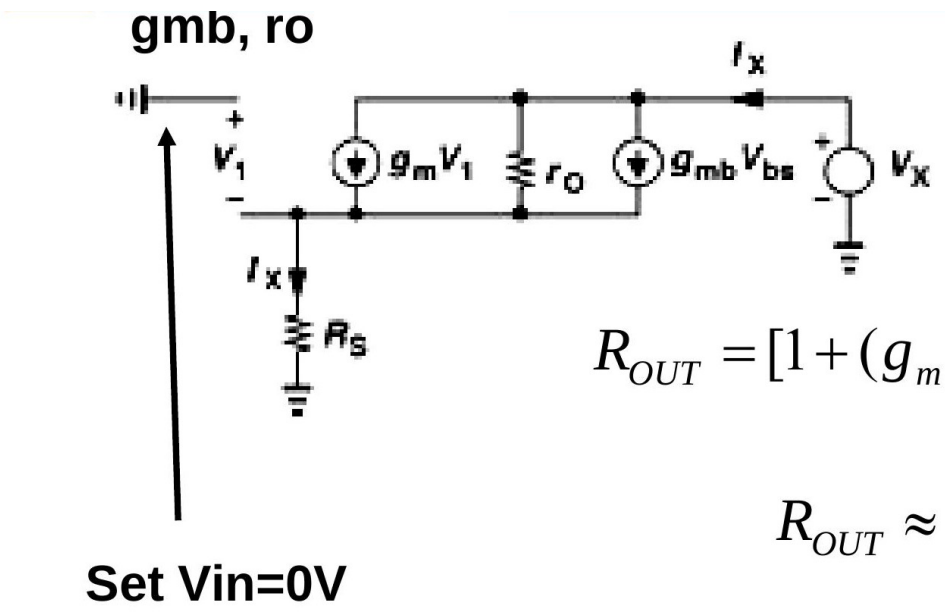
V

Output Resistance of the CS Stage

- Test-source derivation of Rout
- Physical intuition for the RS boost
- Combined Av = −Gm·Rout formula

Rout of the Degenerated CS Stage

- With Vin = 0, a test source Vx/Ix gives Rout = [1+(gm+gmb)ro]RS + ro
- A useful approximation: Rout ≈ ro·gm·RS when gmRS ≫ 1
- The overall gain becomes Av = −Gm(RD∥Rout)



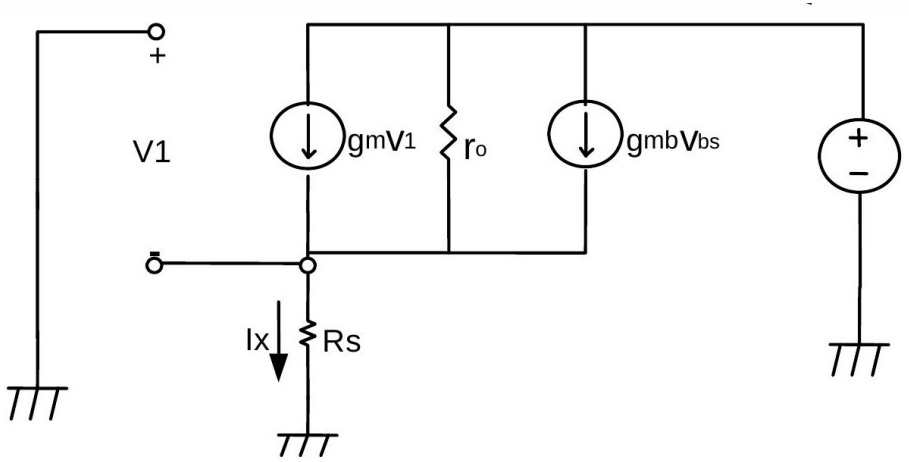
$$R_{OUT} = [1 + (g_m + g_{mb}) r_o] R_S + r_o$$

$$R_{OUT} \approx r_o g_m \cdot R_S$$

$$A_v = -G_m (R_D \parallel R_{OUT})$$

Rout Derivation — Test-Source Method

- Setting V1 = −Vs and Vbs = 0−Vs expresses every controlled source in terms of the single test voltage Vs
- Summing the currents through ro, gmVs, and gmbVs and solving Vx/Ix reproduces Rout ≈ ro·gm·RS



$V_1 = -V_s$

$V_{bs} = 0 - V_s$

$$V_x = r_o\,[I_x + g_m V_s + g_{mb} V_s] + I_x R_s$$

$$V_s = I_x R_s$$

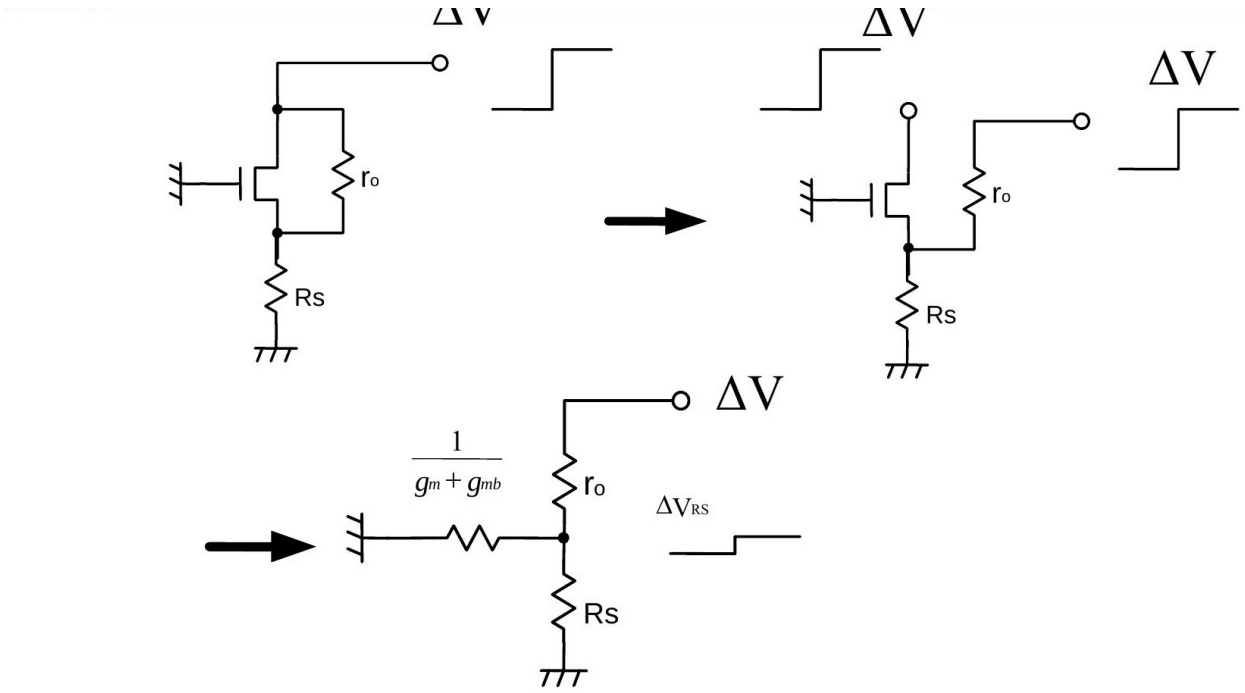
$$V_x = r_o\,[I_x + g_m R_s I_x + g_{mb} I_x R_s] + I_x R_s$$

$$\frac{V_x}{I_x} = r_o + r_o g_m R_s + r_o g_{mb} R_s + R_s = r_o\,[\,1 + (\,g_m + g_{mb}\,)\,R_s\,] + R_s$$

$$\approx r_o g_m \times R_s$$

Why RS Boosts Output Resistance

- A ΔV disturbance at the drain is fed back through RS to modulate the gate-source voltage
- This local feedback opposes the change in drain current, effectively raising Rout
- The DC bias (VDS) itself does not change the current through gm and gmb — only the AC feedback path does



VDS bias does not modify the current through gm and gmb.

Formal ΔV/ΔI Derivation of Rout

- Relating ΔVRS to the applied ΔV through the (gm+gmb) feedback path gives an expression for ΔI
- Taking the limit ΔV/ΔI yields Rout ≈ [1+(gm+gmb)RS]ro + RS, confirming the earlier result algebraically

$$\Delta V_{RS} = \Delta V \frac{\frac{1}{g_m + g_{mb}} \| R_s}{\frac{1}{g_m + g_{mb}} \| R_s + r_o}$$

$$\Delta I = \frac{\Delta V_{RS}}{R_S} = \Delta V \frac{1}{[1 + (g_m + g_{mb})R_s]r_o + R_s]}$$

$$\frac{\Delta V}{\Delta I} = [1 + (g_m + g_{mb})R_s]r_o + R_s]$$

$$\approx R_{out}$$

Full Gain Expression: Av = −Gm·Rout (***)

- Substituting the complete Gm and Rout expressions gives one unified formula for the degenerated-CS gain
- Every degeneration, body-effect, and channel-length-modulation term is captured in a single closed form

$$A_v = -G_m \times R_{out}$$

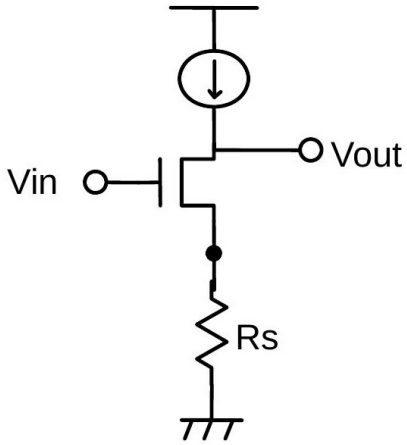
$$= -\frac{g_m r_o}{R_s + [1 + (g_m + g_{mb})R_s]r_o} \times [R_D \| [R_S + r_o + r_o(g_m + g_{mb})R_s]$$

$$A_v = - G_m \times R_{out}$$

Ex. 3.6 — Putting the Formulas to Work

- Using Gm = gm/(1+gmRs) and Rout = ro+gm·ro·Rs, the gain becomes Av = Gm·Rout ≈ gm·ro
- For gmRs ≫ 1, the degenerated stage's gain approaches the same intrinsic-gain limit as the plain CS stage

EX 3.6) calculate the voltage gain



$$A_V = -G_m \times R_{out}$$

$$G_m = \frac{g_m}{1+g_m R_s}$$

$$R_{out} = r_o + g_m r_o R_s$$

$$A_v = G_m R_{out} = \frac{g_m}{1+g_m R_s}\left(r_o + g_m r_o R_s\right) = g_m r_o$$

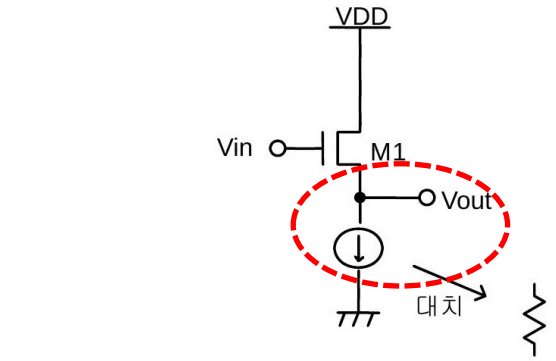
VI

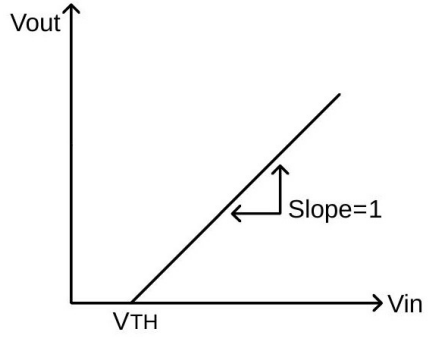
Source Follower (Common Drain)

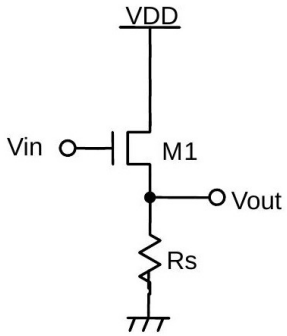
- Gm and Rout derivation
- Three key disadvantages
- Buffering advantage

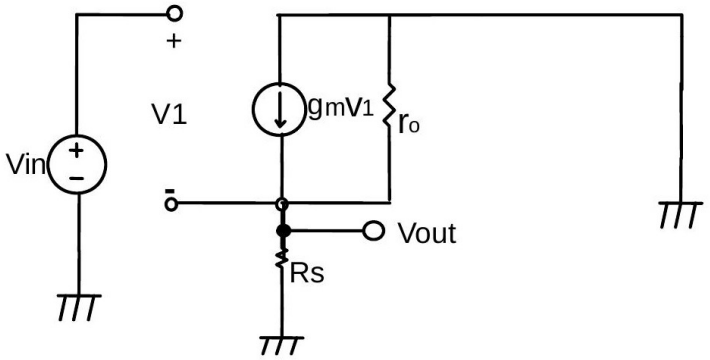
Source-Follower Topology & Transfer Curve

- The output is taken from the source terminal; the drain is tied to a fixed supply
- Once M1 turns on, Vout tracks Vin with an ideal small-signal slope of 1 — it "follows" the input
- RS sets the bias current and, together with gm, the actual (sub-unity) achievable gain



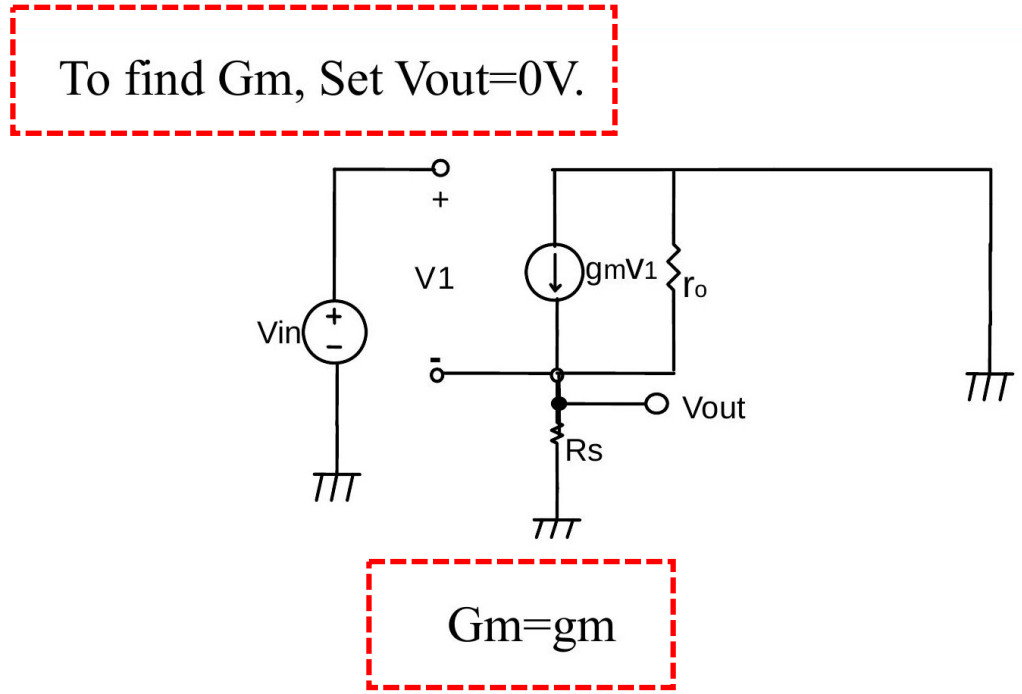






Finding Gm — Set Vout = 0

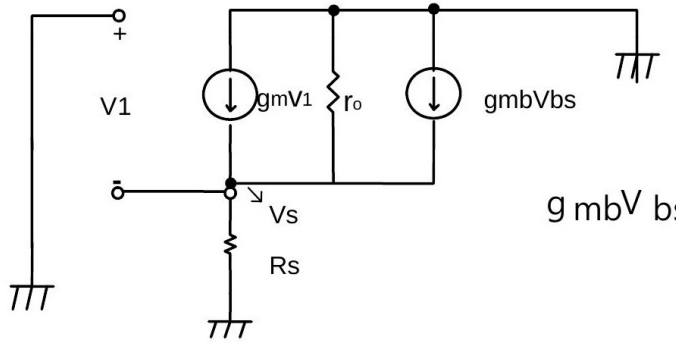
- Shorting Vout to ground makes Vgs = Vin directly, so Gm = gm for the source follower
- This is simpler than the degenerated-CS case since there is no separate load resistor under this test condition



Finding Rout — Set Vin = 0

- With Vin grounded, gmbVbs = gmb(−Vs) = gmb(−Vout) ties the body and source voltages through the feedback loop
- Solving the node equation yields Rout = RS ∥ [1/(gm+gmb)]

To find Rout, Set Vin=0V.

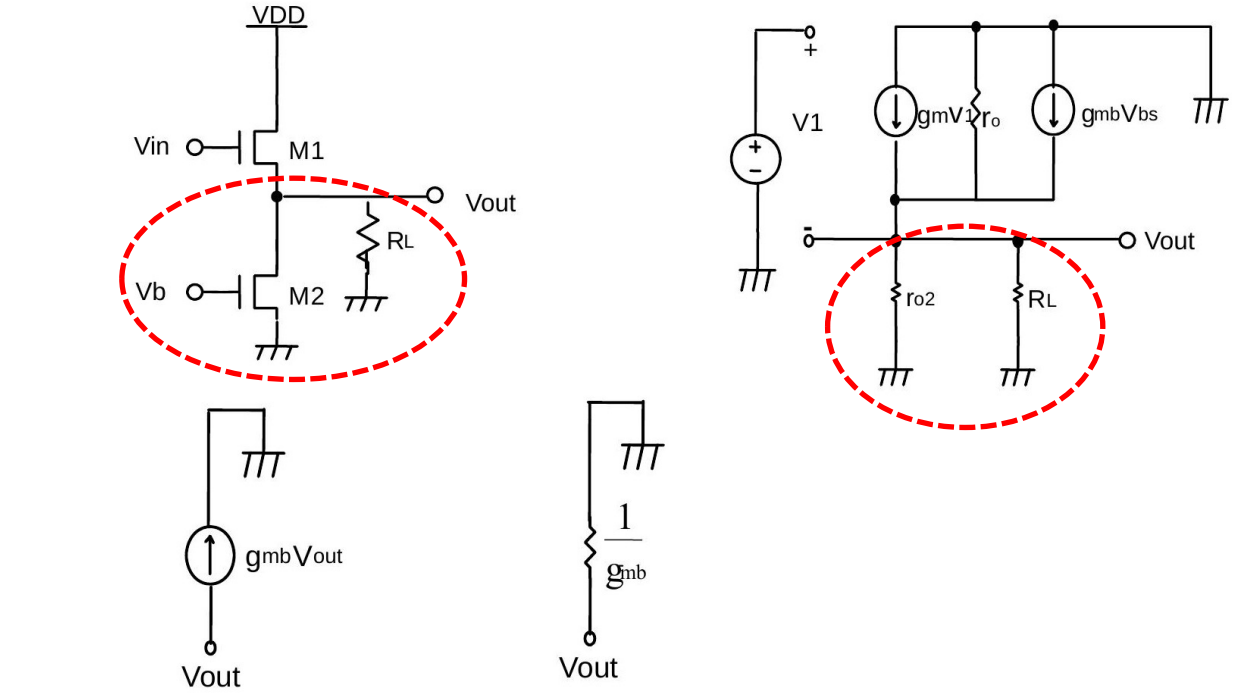


$$g_{mb}V_{bs} = g_{mb}(-V_s) = g_{mb}(-V_{out})$$
$$V_1 = -V_{out}$$
$$G_m V_1 = -g_m V_{out}$$

$$R_{out} = R_S \left\| r_o \right\| \frac{1}{g_m + g_{mb}}$$

Biasing Options: Resistor vs. Current Source

- The tail can be biased with a simple resistor RS or with an ideal current source for improved output resistance
- Each biasing choice trades off headroom, output resistance, and linearity differently



Rout and Gain — Eq. (3.90)

- Rout = ro2 ∥ (1/gm) ∥ (1/gmb) ≈ Rx ∥ (1/gm), combining the loading effects of ro2, gm, and gmb
- |Av| = gm·Rx/(Rx+1/gm) = Rx/(Rx+1/gm); the sign convention of Av deserves care here

$$G_m = g_m$$

$$R_{out} = r_{02} \left\| R_L \right\| r_{01} \left\| \frac{1}{g_m + g_{mb}} = \left(r_{02} \right\| R_L \left\| r_{01} \right\| \frac{1}{g_{mb}} \right) \left\| \frac{1}{g_m} = R_x \right\| \frac{1}{g_m}$$

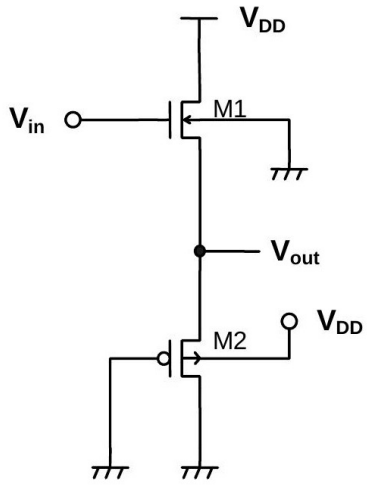
$$|A_V| = g_m \times \frac{\frac{1}{g_m} \times R_X}{R_X + \frac{1}{g_m}} = \frac{R_X}{R_X + \frac{1}{g_m}} \quad \Rightarrow A_V$$

$$= \frac{r_{02} \left\| R_L \right\| r_{01} \left\| \frac{1}{g_{mb}} \right.}{r_{02} \left\| R_L \right\| r_{01} \left\| \frac{1}{g_{mb}} + \frac{1}{g_m} \right.} \quad \Rightarrow \text{Eq 3.90}$$

Ex. 3.8 — Complete Gm, Rout, and Av Calculation

- Gm = gm1, and Rout combines ro1, ro2, and 1/(gm2+gmb2) all in parallel
- The resulting Av expression folds every parasitic resistance into one ratio, giving the realistic sub-unity follower gain

Ex 3.8)



$$G_m = g_{m1}$$

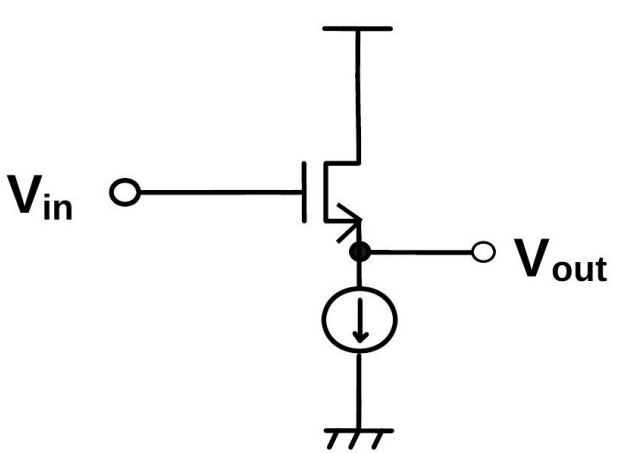
$$R_{out} = \frac{1}{g_{m1}} \| \frac{1}{g_{mb1}} \| r_{o1} \| r_{o2} \| \frac{1}{g_{m2}} \| \frac{1}{g_{mb2}}$$

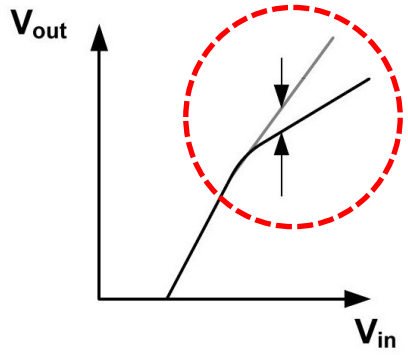
$$A_V = G_m R_{out} = g_{m1} \times \frac{\frac{R_X}{g_{m1}}}{\frac{1}{g_{m1}} + R_X} = \frac{R_X}{\frac{1}{g_{m1}} + R_X} = \frac{\frac{1}{g_{mb1}} \| r_{o1} \| r_{o2} \| \frac{1}{g_{m2}+g_{mb2}}}{\frac{1}{g_{m1}} + \left(\frac{1}{g_{mb1}} \| r_{o1} \| r_{o2} \| \frac{1}{g_{m2}+g_{mb2}} \right)}$$

Disadvantage 1 — Body-Effect Nonlinearity

- As Vout rises, VSB increases, which raises VTH and therefore the VGS needed for the same ID
- Since Vout = Vin−VGS, a Vin-dependent VTH makes the transfer characteristic nonlinear

1.Nonlinearity due to body effect





➢As Vout becomes larger, threshold voltage becomes larger, too

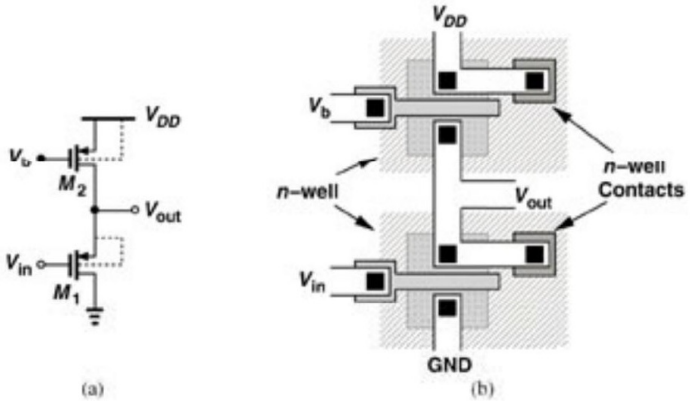
➢Vout=Vin-VGS

➢As VTH becomes larger, VGS becomes larger, too, to keep the same ID

Mitigating Body Effect — Tying Bulk to Source

- Connecting the bulk to the source eliminates body effect and its nonlinearity
- This requires an isolated well for each device, which adds layout area overhead

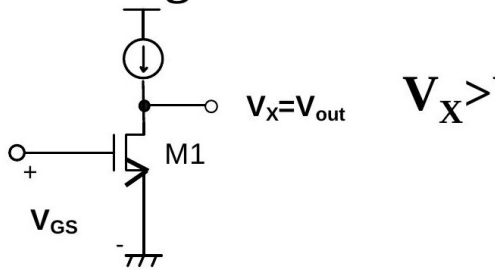
If bulk is tied to the source, body effect can be eliminated
Area overhead is added



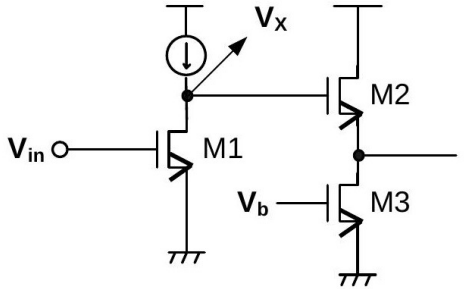
Disadvantage 2 — Voltage Headroom Loss

- Stacking source followers for level-shifting consumes headroom at each stage
- VX > VGS1−VTH keeps M1 in saturation; each additional stacked device tightens this voltage margin further

2.Voltage headroom consumption due to level shift



$V_X > V_{GS1} - V_{TH}$ ⇒ **M1 at the saturation**



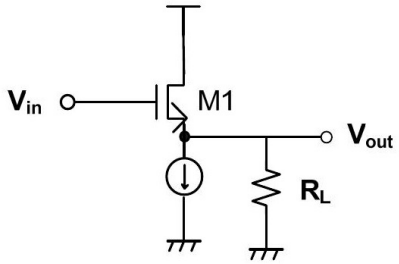
$V_X > V_{GS3} - V_{TH} + V_{GS2}$
⇒ **M3 at the saturation**
⇒ **Headroom margin of V_X reduced**

➢What is a advantage of SF?
=> Small Rout Suitable to Voltage Buffer

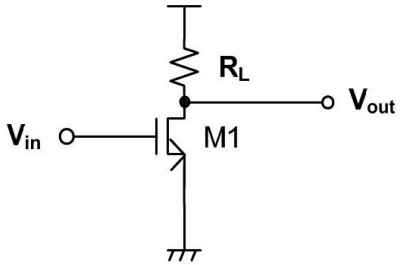
Disadvantage 3 — Limited Drive Capability

- With RL → ∞, the maximum small-signal voltage gain of a source follower is only 1
- Its voltage gain is inherently much lower than a common-source stage biased under the same conditions

3. Poor driving capability



Source Follower



Common Source

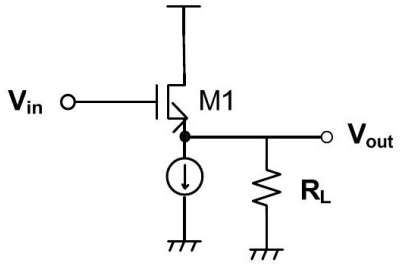
➢When RL is infinite, the maximum voltage gain of SF is 1.
➢Voltage gain of SF much lower than CS amp

Advantage — Low Output Resistance for Buffering

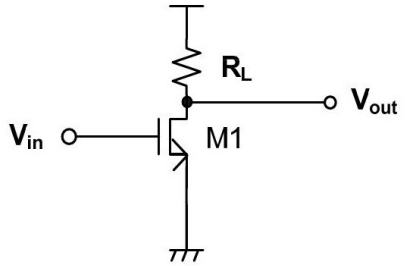
- Despite its gain limitations, the source follower's small Rout makes it well-suited as a voltage buffer
- It can drive a low-impedance load with minimal signal attenuation, unlike a common-source stage

➢What is a advantage of SF?

=> Small Rout Suitable to Voltage Buffer



Source Follower

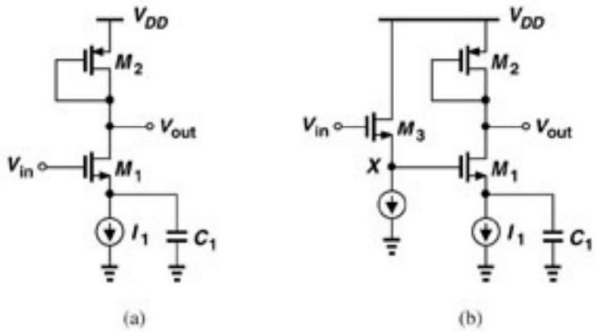


Common Source

Ex. 3.9 — Gain and Saturation Bias Condition

- (a) Av = gm1[ro1‖ro2‖1/gm2] combines every loading path into the follower's small-signal gain
- (b) Keeping M1 in saturation when Vin = VDD requires VGS3−VTH1 > |VGS2|

Ex3.9) (a) calculate the voltage gain of (a)
(b) To leave M1 in saturation, calculate the bias condition, if Vin=VDD



$$(a) A_v = g_{m1}\left[r_{o1} \| r_{o2} \| 1/g_{m2}\right]$$

$$(b) V_{DD} - V_{GS3} + V_{TH1} < V_{DD} - |V_{GS2}|$$

$$V_{GS3} - V_{TH1} > |V_{GS2}|$$

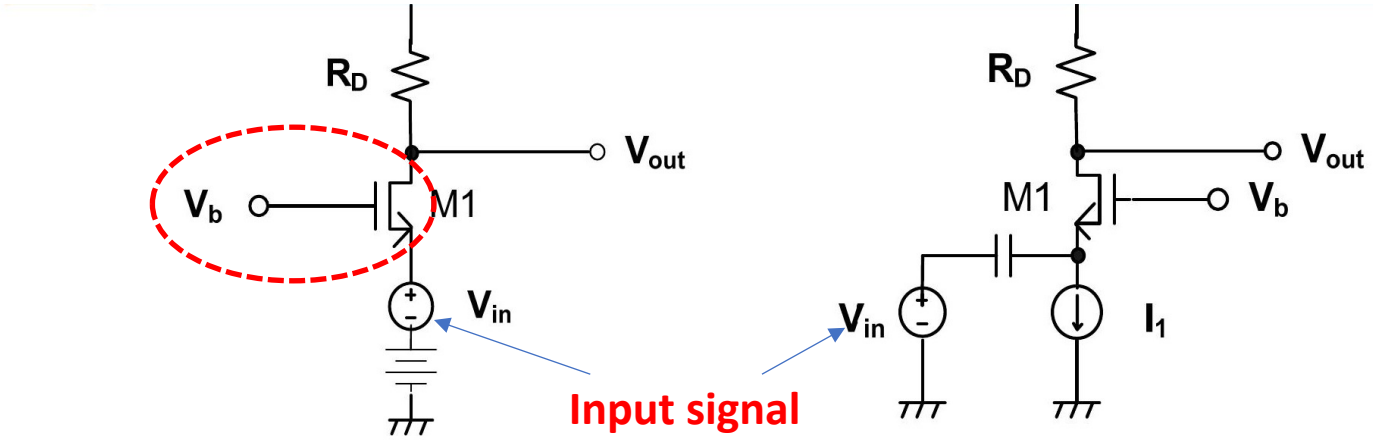
VII

Common-Gate Amplifier

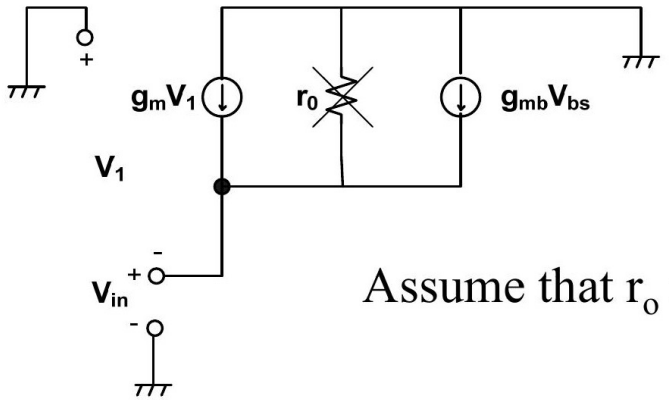
- Gm and Rout derivation
- Input resistance and impedance matching
- Worked design examples

Common-Gate Topology & Finding Gm

- The input is applied at the source, the gate is AC-grounded through a bias Vb, and the output is at the drain
- Setting Vout = 0 with ro assumed infinite gives the basic small-signal gain relationship for this stage



Find Gm
=> Set Vout=0V

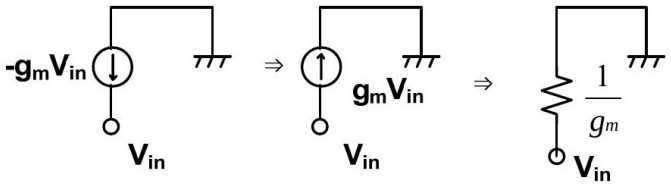


Assume that r_o is infinite

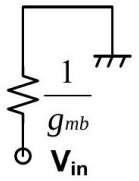
Small-Signal Gm Derivation (ro = ∞)

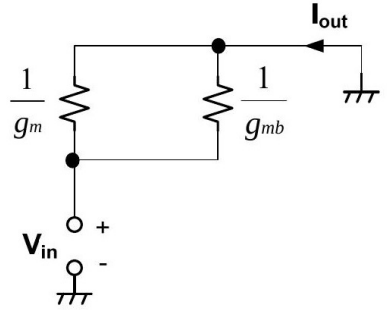
- V1 = −Vin at the source terminal, so both the gm and gmb current sources contribute −(gm+gmb)Vin
- Iout/Vin = −(gm+gmb) — both transconductance mechanisms add directly in the common-gate stage

$$g_m V_1 = g_m \times (-V_{in}) = -g_m V_{in}$$



$$g_{mb} V_{bs} = g_{mb} \times (-V_{in}) = -g_{mb} V_{in}$$





$$V_{in} \times (g_m + g_{mb}) = -I_{out}$$

$$\frac{I_{out}}{V_{in}} = -(g_m + g_{mb})$$

Finding Rout — Set Vin = 0

- With the source grounded and ro assumed infinite, Rout reduces simply to RD
- The voltage gain in this case is Av = −GmRout = +(gm+gmb)RD — note the positive sign: common gate is non-inverting

Find Rout => Set Vin=0V

$$\therefore R_{out} = R_D$$

$$A_V = -G_m R_{out} = +(g_m + g_{mb}) R_D$$

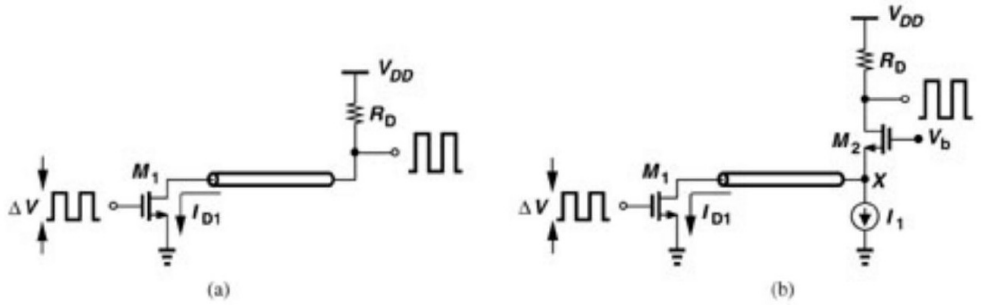
Ex. 3.10 — Gain and Impedance Matching

- (a) With CLM and body effect neglected, the low-frequency voltage gain reduces to Av = gmRD
- (b) Minimizing wave reflection at the input node requires setting 1/gm2 to match the source/line impedance (here, 50 Ω)

Ex3.10) assume channel length modulation=0, body effect=0

(a) Calculate the voltage gain for low frequencies

(b) What condition is necessary to minimize wave reflection at node X?



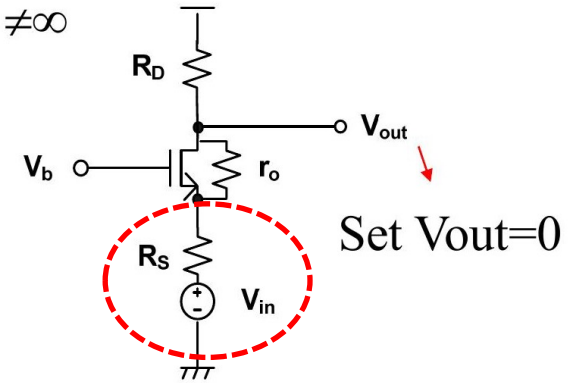
(a) Voltage gain = $g_m R_D$

(b) $1/g_{m2}=50\Omega$

Gm with Finite Source Resistance Rs and ro

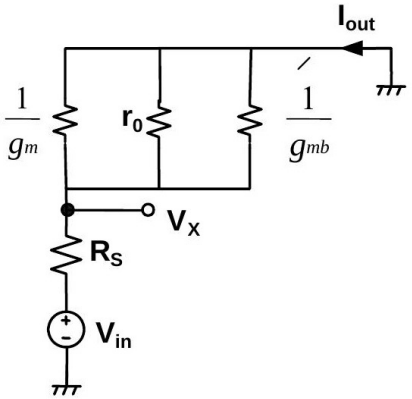
- Including a finite source resistance Rs and finite ro requires solving a full node equation at Vout = 0
- Iout/Vin combines the Rs path with the ro ‖ [1/(gm+gmb)] parallel combination

When $R_S \neq 0$ and $r_o \neq \infty$



➢Find Gm

Set Vout=0



$$V_{in} + I_{out} \times R_s + I_{out} \times \left(\frac{1}{g_m + g_{mb}} \| r_o\right) = 0$$

$$\frac{I_{out}}{V_{in}} = -\frac{1}{Rs + \frac{1}{g_m + g_{mb}} \| r_o} = -\frac{1}{Rs + \frac{\frac{r_o}{g_m + g_{mb}}}{\frac{1}{g_m + g_{mb}} + r_o}}$$

Simplified Gm Expression

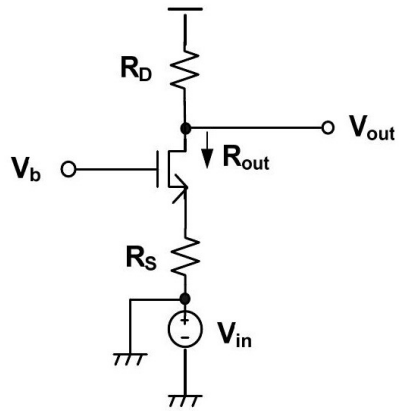
- Combining terms yields Gm = [1+ro(gm+gmb)] / [Rs+Rs·ro(gm+gmb)+ro]
- This single formula captures how source resistance Rs degrades the effective transconductance from the ideal (gm+gmb) value

$$G_m = -\frac{\frac{1}{g_m + g_{mb}} + r_o}{Rs\left(\frac{1}{g_m + g_{mb}} + r_o\right) + \frac{r_o}{g_m + g_{mb}}} = -\frac{1 + r_o(g_m + g_{mb})}{Rs + Rsr_o(g_m + g_{mb}) + r_o}$$

Rout with Finite Source Resistance

- Rout = RD ‖ {[1+(gm+gmb)Rs]ro + Rs}, mirroring the source-degeneration Rout structure
- The Rs term boosts the output resistance the same way source degeneration did in the CS stage

➢Find Rout
➢Set Vin=0



$$R_{out} = R_D \| \{[1+(g_m+g_{mb})Rs]r_o + Rs\}$$
$$= \frac{R_D \times [ro+(g_m+g_{mb})Rsr_o+Rs]}{RD+ro+(g_m+g_{mb})r_oRs+Rs}$$

Voltage Gain — Eq. (3.104)

- Av = −Gm·Rout combines the Gm and Rout expressions from the previous slides into one closed form
- Simplifying gives Av = [RD(1+ro(gm+gmb))] / [RD+Rs+(gm+gmb)roRs] — the general common-gate gain with source resistance

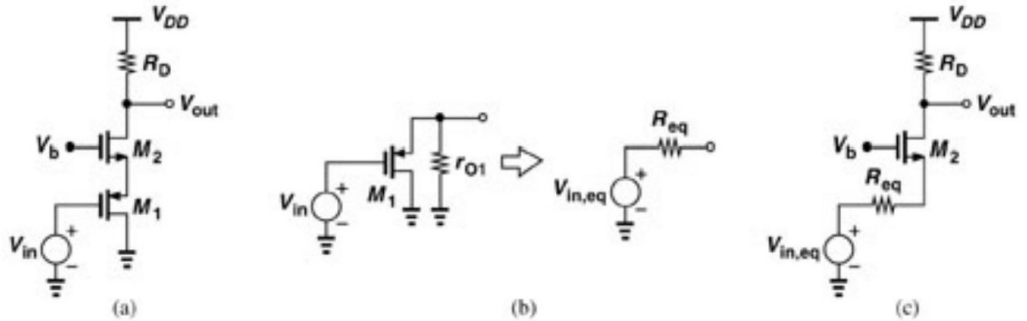
$$Av = -G_m R_{out} = \frac{1+(g_m+g_{mb})r_o}{Rs+(g_m+g_{mb})r_o Rs+ro} \times \frac{R_D \times \left[ro+(g_m+g_{mb})Rsr_o+Rs\right]}{R_D+ro+(g_m+g_{mb})r_o Rs+Rs}$$

$$= \frac{R_D \times (1+r_o(g_m+g_{mb}))}{R_D+ro+Rs+(g_m+g_{mb})r_o Rs} \quad \Rightarrow \text{Eq. 3.104}$$

Ex. 3.11 — Including CLM and Body Effect

- Vin,eq = [ro1‖(1/gmb1)] / [ro1‖(1/gmb1)‖(1/gm1)] · Vin captures the loading from the input source
- Req = ro1‖(1/gmb1)‖(1/gm1) is the resistance the source sees looking into the common-gate input

Ex3.11) calculate the voltage gain, if CLM is not zero and body effect is not zero



$$V_{in,eq} = \frac{r_{o1} \| 1/g_{mb1}}{r_{o1} \| 1/g_{mb1} + 1/g_{m1}} V_{in}$$

$$R_{eq} = r_{o1} \| 1/g_{mb1} \| 1/g_{m1}$$

Ex. 3.11 — Final Gain Expression

- Av = [RD(1+ro(gm+gmb))] / [RD+ro+Rs+(gm+gmb)roRs]
- Rs = Req = ro1∥(1/gmb1)∥(1/gm1), tying the source resistance directly to the input stage's own output impedance

Ex3.11)

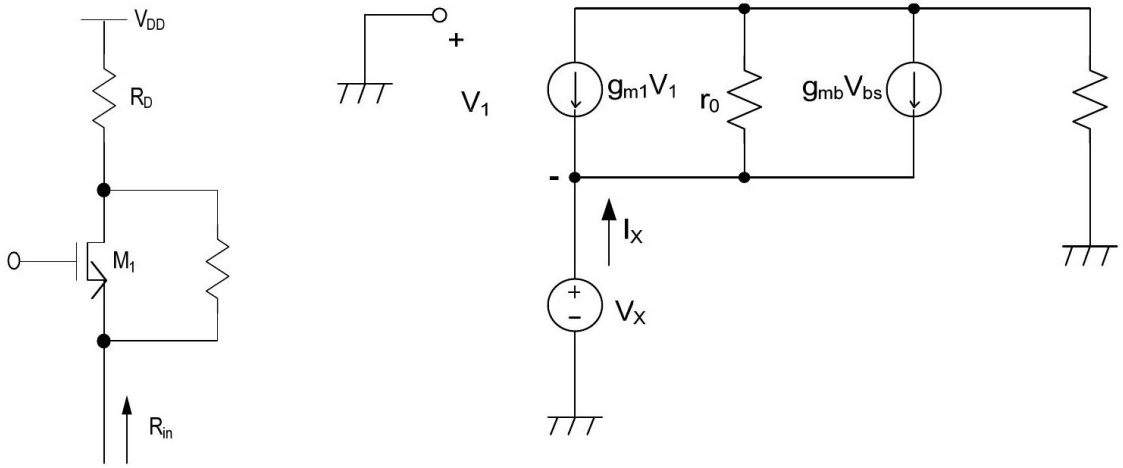
$$\text{Av} = \frac{R_D \times (1 + r_o(g_m + g_{mb}))}{R_D + ro + Rs + (g_m + g_{mb}) r_o Rs}$$

$$R_s = R_{eq} = r_{o1} \,||\, 1/g_{mb1} \,||\, 1/g_{m1}$$

Input Resistance of the Common-Gate Stage

- Unlike CS and source-follower stages, common-gate input resistance is a key design parameter, especially for impedance matching
- The next slides derive Rin by injecting a test current Ix at the source node

➢Input Resistance of CG



Rin Derivation via Test Current

- Writing the node equation at the source (test current Ix, test voltage Vx) gives Rin = Vx/Ix = (ro+RD)/[1+ro(gm+gmb)]
- For large ro(gm+gmb), Rin approximates 1/(gm+gmb) + RD/[ro(gm+gmb)] — set mainly by the transistor's own transconductance

➢Input Resistance of CG

$$V_X = (I_X + g_{m1}V_1 + g_{mb}V_{bs})r_o + I_X R_D, \quad (V_1 = -V_X, V_{bs} = -V_S = -V_X)$$

$$V_X = (I_X - g_m V_X - g_{mb}V_X)r_o + I_X R_D$$

$$V_X(1 + g_m r_o + g_{mb} r_o) = I_X(r_o + R_D)$$

$$\therefore R_{in} = \frac{V_X}{I_X} = \frac{r_o + R_D}{1 + r_o(g_m + g_{mb})}$$

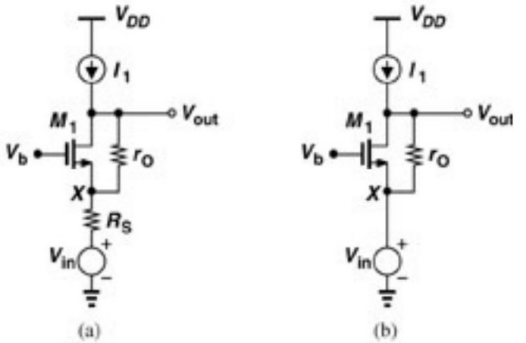
$$\cong \frac{r_o}{(g_m + g_{mb})r_o} + \frac{R_D}{(g_m + g_{mb})r_o}$$

$$\cong \frac{1}{g_m + g_{mb}} + \frac{R_D}{(g_m + g_{mb})r_o}$$

Ex. 3.12 — Voltage Gain with Source Resistance

- Av = [(gm+gmb)/(1+(gm+gmb)RS)] × [ro+(gm+gmb)roRS‖RD] = (gm+gmb)ro
- This matches the same output-boosting pattern seen earlier for the degenerated CS stage

Ex3.12) calculate the voltage gain



$$A_v = \frac{g_m + g_{mb}}{1+(g_m + g_{mb})R_S}(r_o + (g_m + g_{mb})r_o R_s)$$
$$= (g_m + g_{mb})r_o$$

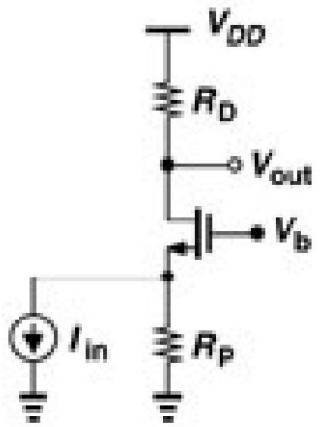
Ex. 3.13 — Current-Drive Gain and Rout

- Driving the stage with a current source Iin instead of a voltage source changes the transfer ratio to Vout/Iin
- Rout combines RP, RD, and the (gm+gmb)ro boosting term from the current-source loading at the drain

Ex3.13) calculate Vout/Iin and Rout

$$\frac{V_{out}}{I_{in}} = \frac{(g_m + g_{mb})r_o + 1}{r_o + (g_m + g_{mb})r_o R_P + R_P + R_D} R_D R_P$$

$$R_{out} = \left[R_P + (g_m + g_{mb})r_o R_P + r_o\right] \| R_D$$



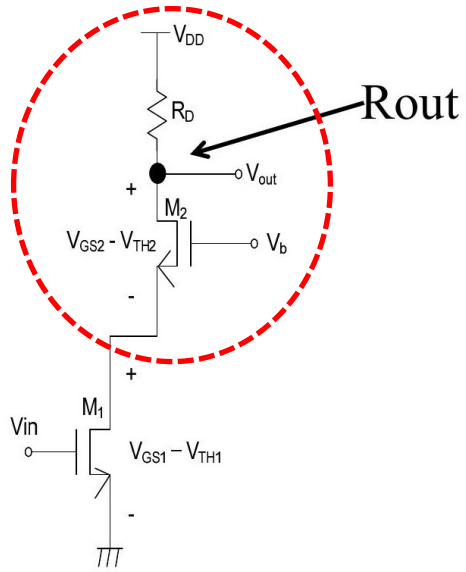
VIII

Cascode & Folded-Cascode

- Rout boosting via cascoding
- Current-source mismatch suppression
- Folded-cascode topology

Cascode Topology — Stacking Amplifier Stages (*)

- Cascode means stacking a common-source and a common-gate device to combine their strengths
- Widely used as an amplifier topology to boost output resistance without stacking more voltage-limited gain stages
- Key question: how does the CS stage's output resistance get reflected/boosted through the CG device?



➢Cascode means stack architecture

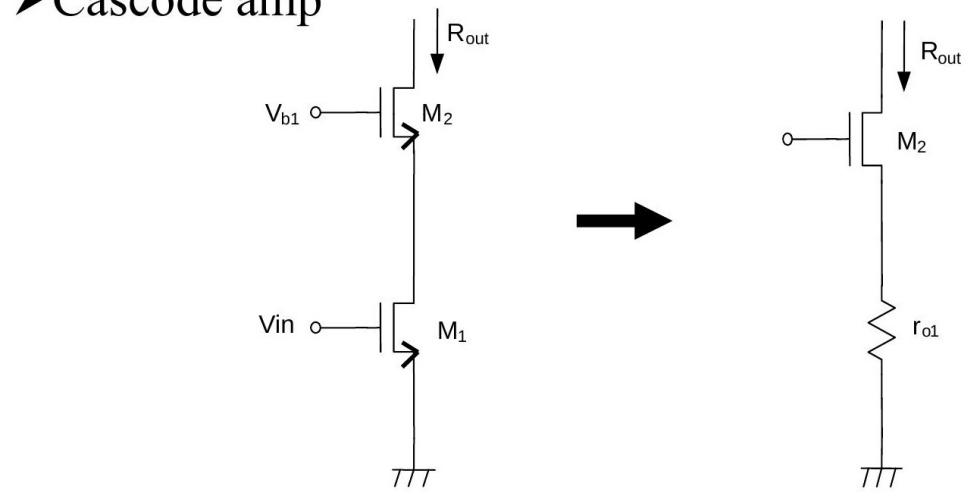
➢Cascode used as a amplifier topology

How is Rout reflected?

Cascode Rout Derivation (*)

- Redrawing the cascode as M1 driving into common-gate M2's source (with Vin = 0 for the Rout test) applies the CG Rout formula directly
- Rout = [1+(gm2+gmb2)ro2]ro1 + ro2 — M1's ro1 is boosted by roughly (gm2+gmb2)ro2

➢Cascode amp



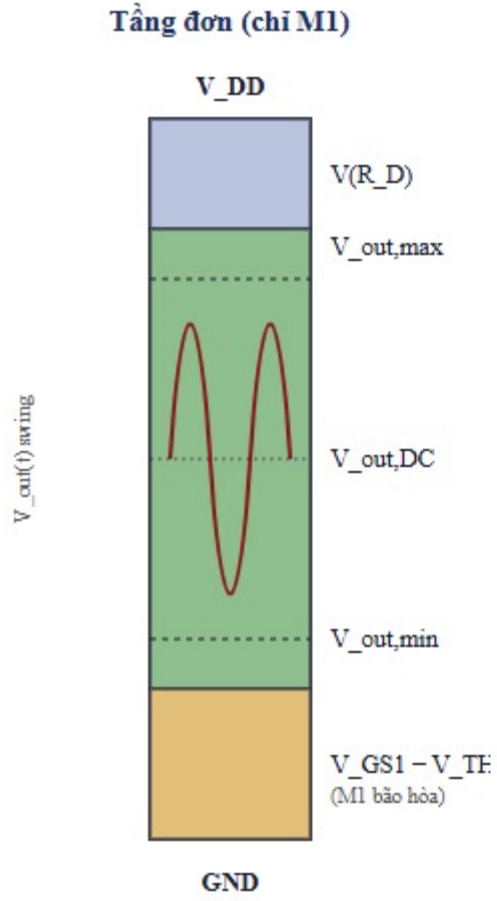
Set Vin=0

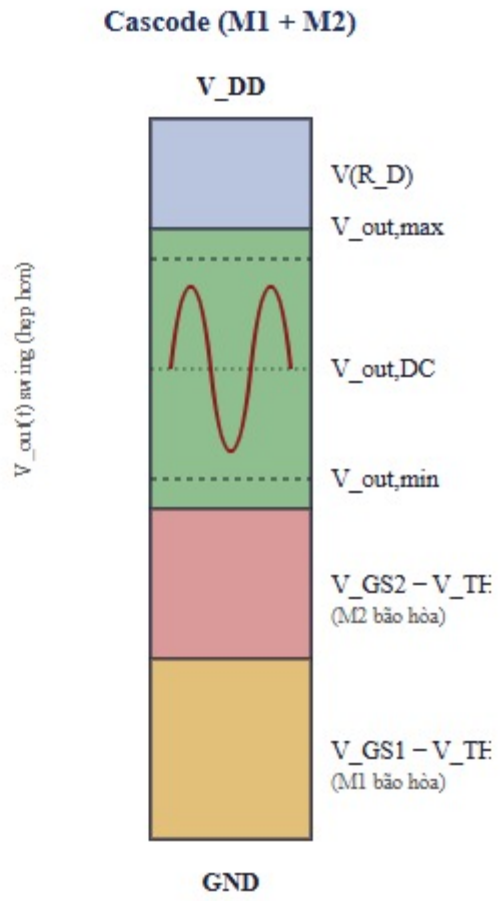
$$R_{out} = [1 + (g_{m2} + g_{mb2})r_{o2}]r_{o1} + r_{o2}$$

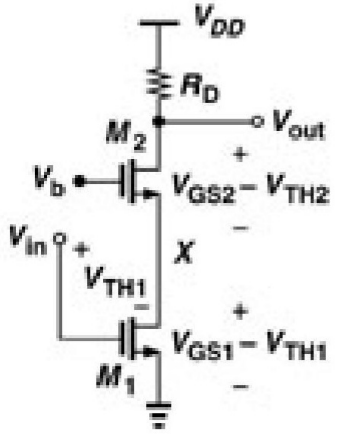
Trade-off: High Rout vs. Reduced Headroom (*)

- Cascoding produces a very large Rout, approaching (gm·ro)·ro — much larger than a single stage
- This benefit comes at the cost of worse DC voltage headroom, since two devices are now stacked in the output branch

Rout is very large, but DC headroom margin is worse

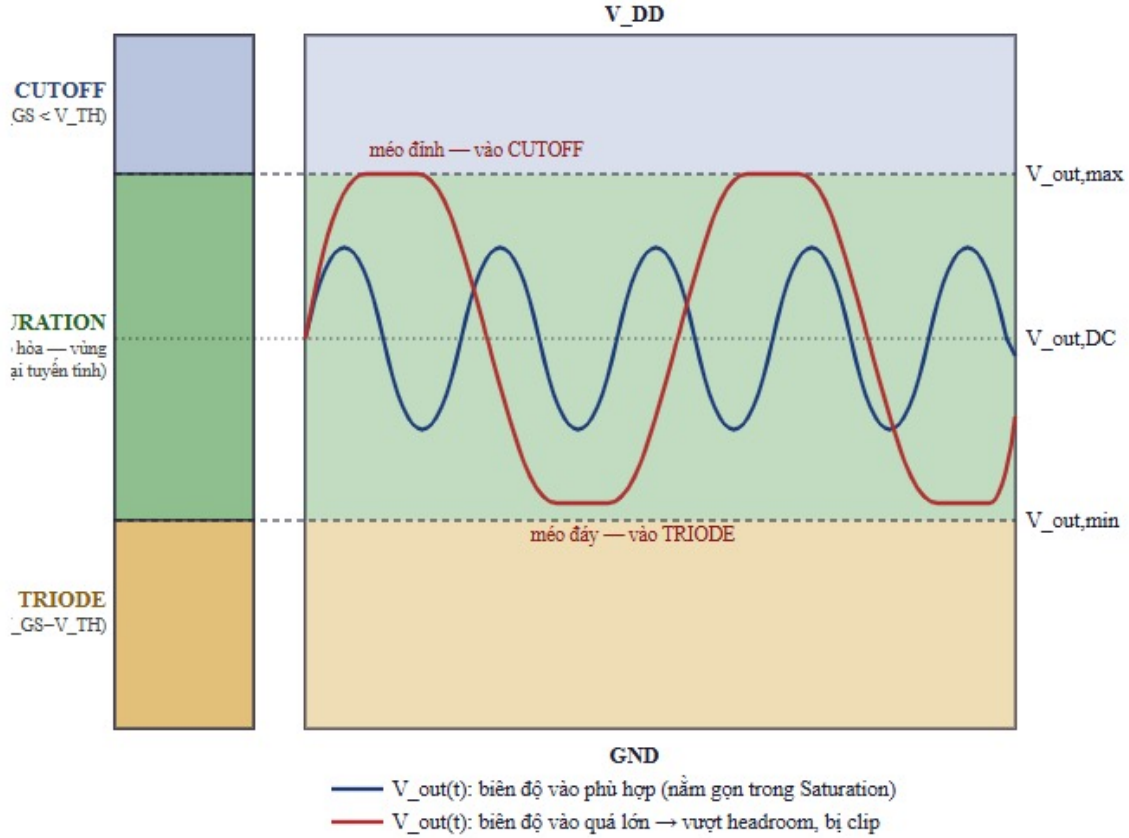






Ba vùng hoạt động của MOSFET và ảnh hưởng lên V_out(t)

Cutoff — Saturation (bão hòa, dùng để khuếch đại) — Triode



Ba vùng hoạt động (theo V_GS, V_DS của transistor):

Vùng	Điều kiện	Hệ quả trên V_out
Cutoff	$V_{GS} < V_{TH}$ (transistor tắt, $I_D \approx 0$)	V_out bị "kẹp" gần V_DD (vì gần như không có sụt áp qua R_D); $g_m \to 0$, mất độ lợi
Saturation	$V_{GS} > V_{TH}$ *và* $V_{DS} \geq V_{GS} - V_{TH}$	Vùng khuếch đại tuyến tính — đây chính là **headroom**: V_out bám tuyến tính theo V_in với độ lợi $A_v = -g_m \cdot R_{out}$
Triode	$V_{DS} < V_{GS} - V_{TH}$	Transistor như một điện trở nhỏ; R_{out} sụp mạnh, độ lợi sụt, V_out bị "kẹp" ở đáy

Khi biên độ tín hiệu vào đủ nhỏ để V_out(t) nằm trọn trong dải Saturation (đường xanh navy), mạch khuếch đại tuyến tính đúng như thiết kế. Khi biên độ vào quá lớn, V_out(t) bị đẩy vượt qua V_out,max hoặc V_out,min, đi vào Cutoff hoặc Triode (đường đỏ) — đỉnh/đáy sóng bị "kẹp phẳng" (clipping), đây là dạng méo phi tuyến kinh điển do vượt quá DC headroom.

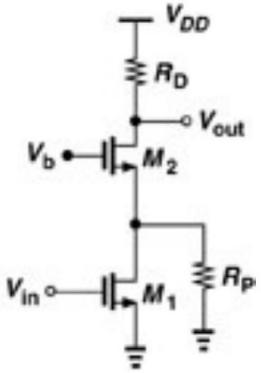
Ex. 3.14 — Voltage Gain (CLM = 0)

- With channel-length modulation neglected, ID2 = gm1Vin·(gm2+gmb2)Rp / [1+(gm2+gmb2)Rp]
- Av = gm1·[(gm2+gmb2)Rp/(1+(gm2+gmb2)Rp)]·RD — M2 passes nearly all of M1's signal current through to RD

Ex3.14) calculate the voltage gain, if CLM=0

$$I_{D2} = g_{m1} V_{in} \frac{(g_{m2} + g_{mb2}) R_P}{1 + (g_{m2} + g_{mb2}) R_P}$$

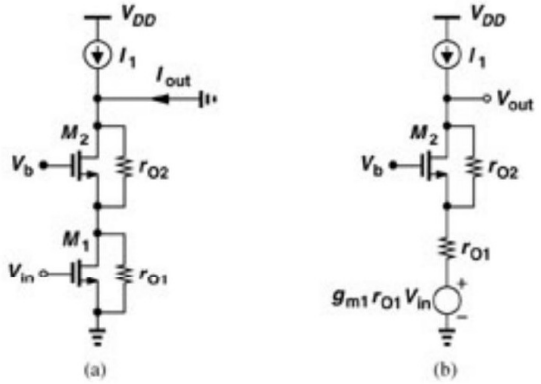
$$A_v = g_{m1} \frac{(g_{m2} + g_{mb2}) R_P}{1 + (g_{m2} + g_{mb2}) R_P} R_D$$



Ex. 3.15 — Exact Voltage Gain

- The exact analysis keeps ro1 and ro2 explicit throughout, giving Gm = Iout/Vin and Av = Gm·Rout
- Av = gm1·ro1[ro2(gm2+gmb2)+1] — the full-precision version of the simplified Ex. 3.14 result

Ex3.15) Calculate the exact voltage gain



$$I_{out} = g_{m1} V_{in} \frac{r_{o1}}{r_{o1} + \dfrac{1}{g_{m2} + g_{mb2}} \| r_{o2}}$$

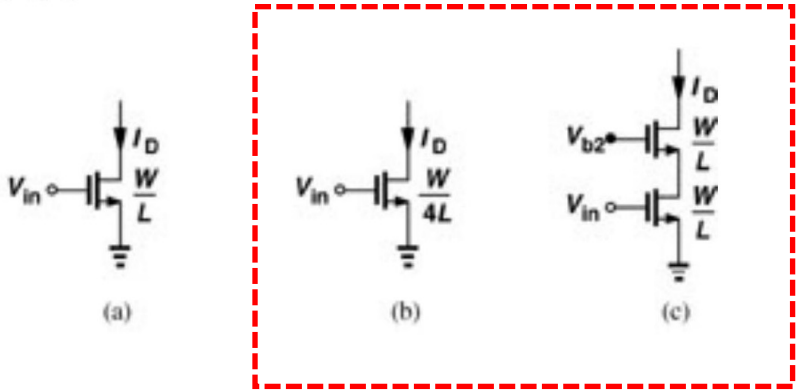
$$G_m = \frac{I_{out}}{V_{in}} = \frac{g_{m1} r_{o1} \left[r_{o2} (g_{m2} + g_{mb2}) + 1 \right]}{r_{o1} r_{o2} (g_{m2} + g_{mb2}) + r_{o1} + r_{o2}}$$

$$A_v = G_m R_{out} = g_{m1} r_{o1} \left[r_{o2} (g_{m2} + g_{mb2}) + 1 \right]$$

Two Ways to Increase Output Resistance

- Increasing channel length raises ro directly, but at the cost of speed, area, and gm
- Cascoding instead multiplies ro by an additional (gm·ro) factor without lengthening the channel

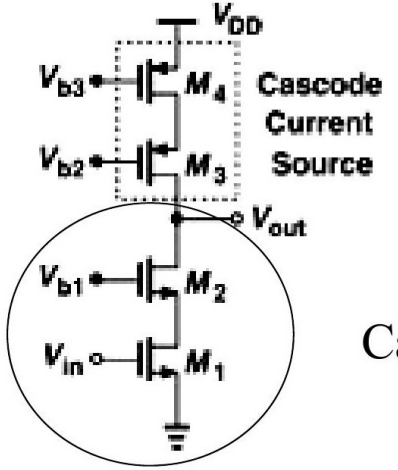
Increasing output resistance by increasing channel length or using cascode



Rout of a Full Cascode Current Source

- Rout = {[1+(gm2+gmb2)ro2]ro1} ‖ {[1+(gm3+gmb3)ro4]ro3}, combining the NMOS and PMOS cascode branches
- |Av| ≈ gm1[(ro1ro2gm2) ‖ (ro3ro4gm3)] follows directly by multiplying this Rout by the input stage's gm1

$$R_{out} = \{[1+(g_{m2}+g_{mb2})r_{o2}]r_{o1}+r_{o2}\} \| \{[1+(g_{m3}+g_{mb3})r_{o3}]r_{o4}+r_{o3}\}$$



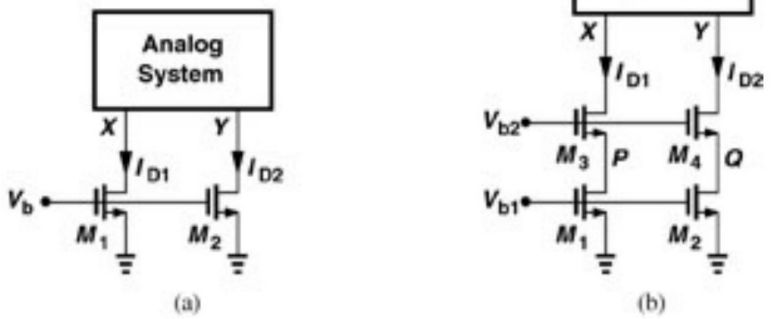
Cascode Amp

$$|A_V| \approx g_{m1}[(r_{o1}r_{o2}g_{m2}) \| (r_{o3}r_{o4}g_{m3})]$$

Ex. 3.16 — Current-Source Mismatch (Setup)

- Two nominally identical NMOS current sources can differ in drain current if their drain voltages VX and VY differ by ΔV
- This mismatch, ID1−ID2, is exactly what a cascode current source is designed to minimize

Ex3.16) Two identical NMOSs are used as constant current source in (a)
Due to internal circuitry, V_X is larger than V_Y by ΔV
(a) Calculate the resulting difference between I_{D1} and I_{D2}



Ex. 3.16 — Current Mismatch, Simple vs. Cascode

- Simple (non-cascoded) source: ID1−ID2 = ½μnCox(W/L)(Vb−VTH)²·(λΔV)
- Cascoded version: the mismatch shrinks by an extra 1/[(gm+gmb)ro] factor, confirming cascoding suppresses current mismatch

For (a)

$$I_{D1} - I_{D2} = \frac{1}{2} u_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 (\lambda V_{DS1} - \lambda V_{DS2})$$

$$= \frac{1}{2} u_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 (\lambda \Delta V)$$

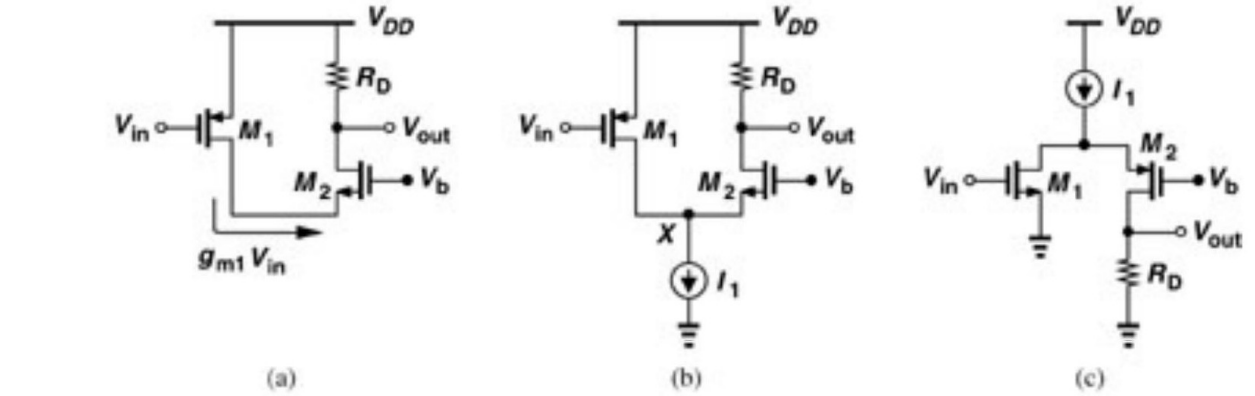
For (b)

$$\Delta V_{PQ} = \Delta V \frac{1}{(g_{m3} + g_{mb3}) r_{o3}}$$

$$I_{D1} - I_{D2} = \frac{1}{2} u_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 \frac{\lambda \Delta V}{(g_{m3} + g_{mb3}) r_{o3}}$$

Folded-Cascode Topology

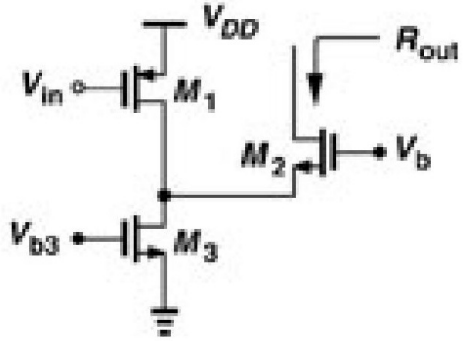
- The folded cascode routes the signal through a complementary (opposite-type) cascode device instead of stacking same-type devices
- This relaxes the headroom penalty of a conventional (telescopic) cascode while preserving the Rout boost



Ex. 3.17 — Folded-Cascode Output Resistance

- Rout = [1+(gm2+gmb2)ro2](ro1∥ro3) + ro2, combining the folded NMOS/PMOS branches in parallel before the cascode boost
- The (ro1∥ro3) term reflects that the folding node now sees two parallel current paths instead of one

Ex3.17) calculate the output resistance



$$R_{out} = [1 + (g_{m2} + g_{mb2})r_{o2}](r_{o1} \parallel r_{o3}) + r_{o2}$$